

Oracle® Mail

Administrator's Guide

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Introduction to Oracle Mail Administration

Oracle Mail is a reliable, scalable, and secure messaging system that uses Oracle Database to store and manage e-mail, voice-mail, and fax messages for your company. Oracle Mail provides message delivery, browser-based clients, and administration utilities by taking advantage of the Oracle core competencies in providing access to, storing, and managing all types of information.

This chapter contains an overview of Oracle Mail flow and of the various tools used to administer Oracle Mail.

This chapter includes the following topics:

- [Overview of Oracle Mail](#)
- [Overview of Oracle Mail Administration](#)

Overview of Oracle Mail

See Also: "Understanding the Oracle Mail Architecture and Functionality" in Chapter 8 of *Oracle Collaboration Suite Deployment Guide* for an overview of Oracle Mail

Troubleshooting and diagnostic information can be found in [Appendix A, "Troubleshooting Oracle Mail"](#).

Overview of Oracle Mail Administration

There are three levels of administration in Oracle Mail, each with a different set of privileges:

- **System Administrator:** Has privileges to administer and configure the entire Oracle Mail system. System administrators have control over the entire system. In addition to being able to shut down the application, system administrators can shut down the computer.
- **Domain Administrator:** Has privileges to administer and configure Oracle Mail within a particular domain. Domain administrators can manage only domain-specific issues. For example, a domain administrator can create and manage users only in whatever domain they control.
- **User:** Has privileges to access, read, and send e-mail and voice-mail messages, and to administer an individual mail account through client-side rules, automated replies, and message management.

This document is intended for system and domain administrators only.

Several tools are available to administrators for Oracle Mail administration. Each tool is used to accomplish different administration tasks, such as user creation, routing control, and system monitoring.

This section includes the following topics:

- [Oracle WebMail Client](#)
- [Oracle Enterprise Manager Application Server Control Console](#)
- [Oracle Enterprise Manager 10g Grid Control Console](#)
- [Oracle Mail Command-Line Utilities](#)

Oracle WebMail Client

Use the Oracle WebMail client to manage Oracle Mail objects, including domains, users, distribution lists, aliases, newsgroups and peer servers, filters and routing of mail, and archiving policies.

Access the Oracle WebMail client at either of the following URLs:

```
http://host_name:port/pls/portal  
http://host_name:port/um
```

Enter the host name of the system upon which Oracle Mail is installed and enter the Oracle Application Server Web Cache port number.

Click the appropriate subtab in the **Administration** tab of the Oracle WebMail client to perform domain, user, list, alias, news, and policy management tasks.

While the **Administration** tab is part of the Oracle WebMail client, it is only available to those users who have been granted privileges of system or domain administrator.

See Also:

- [Chapter 2, "Managing Oracle Mail Objects"](#) for information about administering Oracle Mail using the Oracle WebMail client
- [Chapter 4, "Oracle Collaboration Suite 10g WebMail Client"](#) for information about the Oracle WebMail client

Note: End-users can find information about the Oracle WebMail client the End-User Documentation Portal. The End-User Documentation Portal is a set of customizable HTML pages that provide an overview of Oracle Collaboration Suite clients, including the Oracle WebMail client. The Oracle WebMail documentation includes procedures for basic management of e-mail, such as managing distribution lists, filters, and shared folders. The End-User Documentation Portal also includes links to the Troubleshooting and FAQ site on the Oracle Technology Network, as well as Oracle Collaboration Suite end-user tutorials.

Administrators can host the End-User Documentation Portal. They can customize default content, add content, and choose what content to show or hide.

The End-User Documentation Portal is included with the installation CD as a ZIP file.

See Also:

- *Oracle Mail Installation Guide for Solaris Operating System* for End-User Documentation Portal installation instructions

Oracle Enterprise Manager Application Server Control Console

The Oracle Enterprise Manager Application Server Control Console is an administration tool used to administer the various Oracle Mail servers, such as Housekeeper, POP, and SMTP.

To log in to the Oracle Enterprise Manager Application Server Control Console:

1. Open a Web browser and enter the following URL:

`http://host_name:port_number/emd/console`

2. Enter a user name and password.
3. Click the name of the Applications Tier in the **Name** column.
4. Click **Mail Application** in the **System Components** section to display the Mail Application page.

Listed in the **Service Targets** section are all of the Oracle Mail servers. Click any one to display the server home page and to manage the chosen server.

See Also: [Chapter 3, "Oracle Mail Servers"](#) for information about administering Oracle Mail servers using Oracle Enterprise Manager Application Server Control Console

Oracle Enterprise Manager 10g Grid Control Console

With Oracle Enterprise Manager Grid Control Console, you can monitor the performance of Oracle Mail and drill down to analyze specific performance metrics. The Grid Control Console provides a convenient grouping and dashboard summarization of Oracle Mail status and performance.

See Also: [Chapter 7, "Monitoring Oracle Mail"](#) for information about administering Oracle Mail using Oracle Enterprise Manager Grid Control Console

Oracle Mail Command-Line Utilities

An administrator can employ various utilities from the command line to perform administrative tasks.

See Also: [Appendix D, "Oracle Mail Command-Line Reference"](#) for a list of Oracle Mail command-line utilities

Managing Oracle Mail Objects

This chapter contains information necessary for managing Oracle Mail objects, including domains, users, distribution lists, newsgroups, and aliases. Additionally, it discusses the delegation of certain administration tasks.

Both system and domain administrators have the necessary privileges to manage Oracle Mail objects, although domain administrators are limited to specific domains.

This chapter includes the following topics:

- [Managing Oracle Mail Domains](#)
- [Managing Oracle Mail Users](#)
- [Managing Distribution Lists](#)
- [Managing Aliases](#)
- [Managing News Objects](#)

Managing Oracle Mail Domains

Domains identify e-mail addresses as being from a specific company. E-mail **domains** can have subdomains that can be administered separately even if they are on the same system. The advantages to administering subdomains on the same system separately are:

- Convenience in accommodating subdomains with different maintenance schedules, which is typical for subdomains in different geographic regions
- Ease of administering subdomains with different default attributes, which is common for subdomains belonging to different organizations

Using the Oracle WebMail client, an administrator can perform domain management tasks, such as modifying default attributes for new users, managing domain settings, and creating domains.

Note: Only system administrators can create domains.

This section includes the following topics:

- [Creating Domains](#)
- [Modifying Domain Settings](#)

Creating Domains

Oracle Mail domains are created using the Oracle WebMail client. The base domain is created automatically during installation of the infrastructure. E-mail domain names can be different than the base domain. For example, you can create e-mail subdomains of `company.com` named `a.company.com`, `b.company.com`, and `c.company.com`.

To create additional domains:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab. The **Domain** subtab displays by default.
3. Click **Create Domain**.

Note: If you do not have system administrator privileges, this link will not display in the user interface (UI).

4. Click the name of the installation for which you want to create the new domain.
5. Select a domain from the **Parent Domain** list.
6. Enter the new domain name in the **Domain** field.
7. Click **Create Domain** to display the Domain Settings page.
8. Click **Submit** to display a confirmation page.

Modifying Domain Settings

Note: Preferences modified for a domain apply only to entries created after the modifications. For example, if the default mail quota for the `oracle.com` domain is changed to 60 megabytes (MB), only users newly created in that domain have the new 60 MB quota. Existing users in that domain retain their old mail quota.

To modify domain settings:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab. The **Domain** subtab displays by default.
3. Click **Domain Settings**.
4. Select an installation from the **Installation** list.
5. Select a domain from the **Domain** list.
6. Click **Submit**.
7. Modify domain attributes in the following fields, as necessary:

- **Location in Public Namespace:** A valid distinguished name within the list server to which distribution lists are synchronized. This specifies the distinguished name of the **Lightweight Directory Access Protocol (LDAP)** container in Oracle Internet Directory, which contains all distribution lists in a public namespace for client lookup.

For example:

```
cn=List,dc=foo,dc=com,cn=um_
system,cn=EMailServerContainer,cn=Products,cn=OracleContext
```

See Also: ["Enabling Distribution List Lookup from Standard Clients"](#) on page G-2 for more information about distribution list synchronization

- **Objectclasses for Creation in Public Namespace:** Enter a list of LDAP objectclasses used while creating distribution lists in the public namespace. The list of LDAP objectclasses must include the groupofnames or groupofuniquenames parameter.

For example:

```
groupofuniquenames=domainRelatedObject
```

- **Naming Attribute for Creation in Public Namespace:** Contains the naming attribute to be used while creating distribution lists in the public namespace.

For example:

```
domain
```

- **Flashback Mail Recovery:** Enables or disables **flashback mail recovery** for e-mail users. This attribute does not affect the flashback mail recovery capability for administrators using the PL/SQL interface.

8. Click **Submit** to apply the changes.

Managing Oracle Mail Users

Using the Oracle WebMail client, you can perform user management tasks, such as adding, removing, and modifying e-mail users.

Notes:

- After a successful installation of Oracle Mail, a user with full administrative privileges (a system administrator) must be created using Oracle Delegated Administration Services prior to any user being provisioned or deprovisioned using Oracle WebMail.
 - Base users having multiple mail IDs cannot be managed from the centralized provisioning framework. Users that have multiple mail IDs cannot be managed from Oracle Delegated Administration Services; Oracle WebMail must be used to manage such users.
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-

See Also: ["oesucr"](#) on page D-11 for information about managing users from the command line

This section includes the following topics:

- [Adding E-mail Users](#)
- [Modifying E-mail User Attributes](#)
- [Setting Default New User Attributes](#)
- [Removing E-mail Users](#)
- [Recovering Deleted Messages](#)

Adding E-mail Users

Notes:

- A base user must exist in Oracle Internet Directory before an e-mail account can be created. If the intended e-mail user has no entry in the directory, a message displays with a link directing you to the directory's Oracle Delegated Administration Services page. You can create and provision the user there.
 - Once users are created using Oracle Delegated Administration Services, it is not necessary to re-create them using the Oracle WebMail client.
-
-

To add e-mail users:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **User** subtab. The Add E-mail Service to User page displays by default.
4. Select a domain from the **E-mail Domain** list.
5. Click **Select** to display the following list of user attributes for an e-mail account:
 - **E-mail User Account Name:** Enter an account name for the user, such as `firstname.lastname`.

Note: Only US-ASCII characters are accepted in this field. Spaces in account names are converted to periods (.).

Account names containing the following non-US-ASCII characters are not supported:

ä
Ä
ö
Ö
ü
Ü
ß

- **Base Subscriber:** Select a domain from the list.

This is the Oracle Internet Directory realm under which the base users are provisioned. The Oracle Internet Directory realm can be different than the domain of the e-mail address. For example, base users can be provisioned under the realm `dc=foo,dc=com`; however, the domain of the e-mail address could be `us.foo.com` or `sales.foo.com`. Use this field to select the Oracle Internet Directory realm under which the base user is provisioned to enable Oracle WebMail to ascertain with which base user to associate this user ID.

- **E-mail Quota (MB):** Enter a number to represent the number of megabytes (MB) of e-mail storage assigned to this user.

1048576 MB is the maximum quota that can be specified in this field.

Enter 0 to give a user unlimited quota.

- **Role:** Specifies the permissions of a particular user.

See Also: ["Overview of Oracle Mail Administration"](#) on page 1-1 for information about the various administrator roles

- **Archive Policy:** Select an [archive policy](#) to assign to the user.

Archive policies can be applied to any e-mail user so that all of the user's incoming and outgoing e-mail messages are stored in a third-party storage facility. If you do not want to associate the user with an archive policy, leave this option blank. If archive policies are created later, you can modify archive policy application for a user.

If no archive policies exist, this field will not be displayed.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about archive policies

6. Click **Add** to add the user to the domain or **Cancel** to return to the previous page.

A Word About Quota

Two [quota](#) values can be set for users: `user-quota` and `voice-quota`. When usage reaches `user-quota`, it means that the sum of e-mails and voice mails is equal to the `user-quota` value. The `voice-quota` is an additional buffer provided to users so that voice-mail delivery is not affected when users reach their quota.

All e-mails and voice mails are delivered to users as long as they are under `user-quota`. Once users reach their `user-quota`, e-mails are held in the system and not delivered. In addition, users cannot save new messages in the server folders when they reach `user-quota`. For example, saving a copy of outgoing messages to the Sent folder is not allowed. The IMAP server informs the client that the user is over quota when the user tries to save new outgoing mail.

However, voice-mail delivery continues as long as total usage is under the `user-quota` plus `voice-quota` value. For example, if `user-quota` is 50 MB and `voice-quota` is 20 MB, e-mail delivery stops after the usage is 50 MB, but voice-mail delivery continues until `user-quota` and `voice-quota` reach 70 MB.

E-mail messages are received as usual when a user is over quota. They are stored in exactly the same tables but there is a flag that sets the status of the message so that it cannot be viewed by the user. The system will continue to receive messages until the disk space fills to capacity.

The message ID of any message received while a user is over quota is set to a negative number to indicate that it was received while the user was over quota. Once the user is

back under quota, the message ID is set to a positive number and the messages can then be retrieved.

When users clean up their accounts and are under the `user-quota` and `user-quota` plus `voice-quota` values, e-mail and voice-mail delivery resume.

Modifying E-mail User Attributes

To modify the attributes of an existing user:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **User** subtab.
4. Click **Modify User**.
5. Enter the user ID in the **Search Criteria** field. Enter an asterisk (*) to perform a wildcard search.
6. Select the user's domain from the **Domain** list.
7. Click **Go**.
8. Click the name of the user in the **E-mail User Account Name** column to modify attributes for that user.
9. Modify the following attributes, as necessary:
 - **E-mail User Account Name:** The e-mail user account address. This information cannot be modified.
 - **Collaboration Suite Database:** Global database name of the information storage database containing this user's folders and messages. This information cannot be modified.
 - **E-mail Quota (MB):** This is the maximum amount of e-mail messages, measured in megabytes, that an individual can store on the mail store. If you enter 0, a user has unlimited quota.
 - **Additional Voice Mail Quota (MB):** This is the maximum amount of voice mail messages, measured in megabytes, that an individual can store on the mail store. If you enter 0, a user has unlimited quota.

This quota is in addition to the e-mail quota. For example, if a user has 40 MB of e-mail quota and 10 MB of voice mail quota, the user has a total of 50 MB of message storage. While e-mail quota cannot exceed the amount given, voice-mail quota can exceed its given amount up to the total amount of message storage.

If either of these quotas is set to 0, a user has unlimited overall storage. However, if the voice-mail quota is set to 0, and e-mail quota is set to a finite value, such as 40 MB, when the voice-mail quota reaches 40 MB, the user will no longer be able to receive e-mail.

- **User State:** Choose either `Active` or `Inactive` from the **User State** list.

Notes:

An inactive user cannot access the mail system at all. However, messages sent to an inactive account are stored in the database and can be accessed if the account is reactivated.

If the user state is changed to inactive and if the user owns mailing lists or newsgroups, Oracle WebMail displays a warning message listing the names of distribution lists or newsgroups owned by the user. This warning message informs the administrator that because the owner of these lists or newsgroups is inactive, some administrative activities handled by the owner will not be possible for these distribution lists or newsgroups.

- **Auto Reply Mode:** Select from the list one of the following automatic reply choices:
 - Reply with original message, along with "Auto Reply Message" below
 - Reject sender's e-mail
 - Vacation, reply only one e-mail with each sender
 - Reply each incoming e-mail with "Auto Reply Message" below
- **Auto Reply Text:** Enter any text to be included in automatic replies.
- **Auto Reply Expiration:** Enter a date at which automatic reply will cease.
- **Forward E-mail Address:** Specify an address to which messages will be forwarded.

If an e-mail address is entered here, an `orclforwardaddress` entry is created in Oracle Internet Directory, and all messages are forwarded to the specified address. No messages are delivered to the original recipient's Inbox.

In order for messages to continue being delivered to the user's Inbox, a filter must be created to accomplish this task.

- **Text Indexing:** When enabled, Oracle Mail servers index messages as soon as possible after the messages enter the system so that the user can search the database and return results in seconds. This task is performed by the Housekeeper server.

Note: Formally known as index synchronization, text indexing can be performed for messages already present in a user's mailbox. Set the `oracle.mail.admin.policy.indexoldmessages` parameter to `true` in the `oc4j.properties` file located in the `$ORACLE_HOME/j2ee/OC4J_OCSCClient/config` directory on each Applications tier. Restart Oracle WebMail after making this change. By default this parameter value is `false`, so text indexing is done only for new messages.

See Also: ["Configuring Housekeeper Tasks"](#) on page 3-24 for more information about index synchronization

- **Role:** Specifies the permissions that a particular user has:

System Administrator: A system administrator will have permissions to administer and configure the entire Oracle Mail system.

Domain Administrator: A domain administrator will have permissions to administer and configure Oracle Mail within a particular domain.

User: A user will have permissions only to access, read, and send e-mail, fax, and voice-mail messages.

- **Archive Policy:** Select an archive policy to assign to the user.

Archive policies can be applied to any e-mail user so that all of the user's incoming and outgoing e-mail messages are stored in a third-party storage facility. If you do not want to associate the user with an archive policy, leave this option blank. If archive policies are created later, you can modify archive policy application for a user.

If no archive policies exist, this field will not be displayed.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about archive policies

10. Click **Modify** to apply the changes.

Setting Default New User Attributes

All new e-mail users created in a particular domain are given default attributes that can later be changed.

To set the default attributes of new users in a particular domain:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab. The Default New User page in the **Domain** subtab displays by default.
3. Select an installation from the **Installation** list.
4. Select a domain from the **Domain** list.
5. Click **Submit**.
6. Modify the following attributes:
 - **E-mail Quota (MB):** Enter a number to represent the number of megabytes of e-mail storage assigned to each user.
Enter 0 to give a user unlimited quota.
 - **Additional Voice Mail Quota (M):** Enter a number to represent the number of megabytes of voice-mail storage assigned to each user.
 - **User State:** Defines the user as active or inactive. If **User State** is set to **Active**, the user has full use of the mail protocols that are exposed by the implementation.
A message sent to an inactive user will result in a **delivery status notification (DSN)** being sent to the sender of the message.
 - **Text Indexing:** When text indexing is enabled, e-mail messages are indexed for searching.

- **Role:** Defines the user as either a regular user, a system administrator, or a domain administrator.

See Also: ["Overview of Oracle Mail Administration"](#) on page 1-1 for more information about system and domain administrators

- **Number of Messages Displayed in Oracle WebMail:** Specifies the number of message headers displayed at one time on the Oracle WebMail client. For example, you can specify to display 20 messages at a time.
- **Display Headers (Oracle WebMail):** Select `Full` from the list to display Oracle WebMail headers in detail.
- **Archive Policy:** Select an archiving policy to assign to new users.

Archive policies can be applied to any e-mail user so that all of the user's outgoing and incoming e-mail messages are kept in an archive. If no archive policies exist or you do not want to associate the user with an archive policy, leave this option blank. If archive policies are created later, you can modify archive policy application for a user.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about archive policies

7. Click **Submit** to apply the changes.

Removing E-mail Users

Notes:

- When an e-mail user is removed, any shared folders and public shared folders owned by that user are also deleted; however, user data stored in the database is not removed.
 - If the user being deleted owns a distribution list or newsgroup, the user deletion will not proceed. An error message displays that will contain the names of the distribution lists or newsgroups owned by this user. The administrator must change the ownership of these lists and newsgroups for deletion of the user to succeed.
-

See Also: ["oesucr"](#) on page D-11 for information about removing a user's data from the database

To remove individual e-mail users:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **User** subtab.
4. Click **Remove User**.

5. Enter the user ID in the **Search Criteria** field. Enter an asterisk (*) to perform a wildcard search.
6. Select the user's domain from the **Domain** list.
7. Click **Go**.
8. Select the user you want to remove.
9. Click **Remove** to display a confirmation page.
10. Click **Return to User Management**.

Recovering Deleted Messages

Using this feature, you can recover deleted messages for users using the LogMiner feature of the database.

Note: Recovering deleted messages from the redo log is an expensive operation because it involves searching the redo log file with the LogMiner utility.

The time it takes to recover deleted messages is a function of:

- The size and number of the redo logs being searched
- The number of messages to be recovered
- The size of the MIME message

The greater the number of messages to be recovered, the longer the operation takes. The operation can take from several minutes to several hours, depending upon the preceding factors. If a large number of messages must be recovered, Oracle recommends running LogMiner during off-peak hours.

You must run Housekeeper server before you can recover deleted messages.

To recover deleted messages:

1. Set up LogMiner to recover deleted messages as described in ["Setting Up LogMiner to Recover Mail"](#) on page 3-26.
2. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

3. Click the **Administration** tab.
4. Click the **User** subtab.
5. Click **Recover E-mail** to display the Recover E-mail page.
6. Enter information in the following fields:
 - **E-mail Account Name:** Enter the fully qualified e-mail address of the user for whom you are recovering messages.
 - **Sender:** Enter the fully qualified e-mail address of the sender if the user is searching for deleted messages by sender.

- **Subject:** Enter a string of text to search the Subject field of deleted messages.
- **Date Range From:** Enter a date from which to begin searching messages.
- **Date Range To:** Enter a date at which to end searching messages.

If more than one field is given for a search, the search is performed on all the criteria. For example, if the **Date Range From**, **Date Range To**, and **Sender** fields contain values, the search returns deleted messages where the **Date Range From** is *mm/dd/yyyy* (or *dd/mm/yyyy*, depending upon your locale), the **Date Range To** is *mm/dd/yyyy* (or *dd/mm/yyyy*, depending upon your locale), and the **Sender** is *sender*.

The **Subject** and the **Sender** fields are used to perform regular POSIX and Unicode expression matching.

7. Click **Search** to display a list of recoverable messages.
8. The list of recoverable messages includes the following information:
 - **Sender:** Fully qualified e-mail address of the sender of the message
 - **Date:** Date that the message was sent
 - **Subject:** Contents of the **Subject** field of the message
 - **Database Instance ID:** The system identifier (SID) of the Oracle instance (especially applicable in an Oracle Real Application Clusters environment) on which the message was deleted, and identifies the redo log file name from which the deleted message must be recovered
 - **Redo Log Number:** The redo log number identifies the archived redo log file from which the deleted message is to be recovered.
 - **Available on Disk?:** If checked, the redo log file from which messages are recovered is online. If it is offline, the redo log file cannot be read and the messages cannot be recovered.

Note: If the **Database Instance ID** and **Redo Log Number** both have a 0 value for a message, and the **Available On Disk?** column is checked for that message, this message can be recovered. In this case, the 0 values indicate that this message has not yet been through the garbage collection process of the Housekeeper server.

9. Select any number of messages and click **Recover**. The same message can be recovered multiple times if the message appears in the recoverable messages list based on the search criteria.
10. Click **OK** when the confirmation page displays. Recovered messages return to the user's inbox. If an error page displays, no messages were recovered. If a warning page displays, some messages were recovered, while others were not.

The SMTP server in the Oracle Internet Directory service registry is used to deliver the recovered messages to the user's Inbox. The received date of the recovered message is changed to the date that the recovered message is received.

Managing Distribution Lists

Distribution lists are managed using the Oracle WebMail client. Management tasks include:

- Create, delete, or show [distribution lists](#) or list members
- Modify distribution list properties
- Show all distribution lists to which a member belongs

Additionally, routing control of incoming mail and certain List Server parameters configurable using Oracle Enterprise Manager Application Server Control Console can be used to further manage how messages are delivered to distribution lists. In some cases, distribution lists can contain many hundreds or even thousands of users, which could potentially interfere with e-mail service.

Limits to the number of recipients in an e-mail envelope (this is applicable to all e-mail traffic, not just messages sent to distribution lists) can be established, in addition to batching recipients and setting a number of e-mails that can be processed concurrently.

See Also:

- ["Configuring Advanced Routing Control Settings for Incoming Mail"](#) on page 8-16
- ["Oracle Mail List Server"](#) on page F-25 for a list of List Server parameters

This section discusses various management tasks and provides distribution list attributes and parameters.

See Also: ["oesdl"](#) on page D-2 for information about managing distribution lists from the command line

This section includes the following topics:

- [Setting Default Distribution List Attributes](#)
- [Distribution List Parameters](#)
- [Creating a Distribution List](#)
- [Modifying Distribution List Properties](#)
- [Modifying Default Distribution List Parameter Values](#)
- [Deleting a Distribution List](#)
- [Managing Distribution List Members](#)
- [Showing All Distribution Lists to Which a User Subscribes](#)

Setting Default Distribution List Attributes

Distribution list attributes include:

- Group type, described in [Table 2-1](#)
- Subscription type, described in [Table 2-2](#)
- Posting type, described in [Table 2-3](#)

To set the default attributes of new distribution lists in a particular domain:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click **Default New List** in the **Domain** subtab.
4. Select an installation from the **Installation** list.
5. Select a domain from the **Domain** list.
6. Click **Submit**.
7. Modify the following attributes, described in [Table 2-4](#):
 - **Collaboration Suite Database**
 - **Owner**
 - **Topic**
 - **Editor's List**
 - **Moderator's List**
 - **Auto Reconfirm**
 - **Group Type**
 - **Group Subscription Type**
 - **Post Type**
 - **Maximum Message Size**
 - **Invite Text**
 - **Information Text**
 - **Merge Tag**
 - **Edit/View Member Privilege**
 - **Enable Digests**
 - **Enable Bounce Processing**

Note: For the **Group Type**, **Group Subscription Type**, **Post Type**, **Edit/View Member Privilege**, **Enable Digest**, and **Enable Bounce Processing** parameters, in a domain hierarchy, if any of these parameters are set in any parent domain, and the child domain does not have any values set, the value of the attribute is inherited from the parent domain.

For example, if the `foo.com` domain has **Edit/View Member Privilege** set as `Members`, `Owner`, `Provers`, `Moderators`, `Editors`, and if `us.foo.com`, a child domain of `foo.com`, does not have a value set for this parameter, Oracle WebMail displays the settings of the parent domain, if available.

If you clear the value in `us.foo.com` by selecting the blank option from the **Edit/View Member Privilege** list and save your settings, the next time you view, you will see that the settings for this parameter are those of the parent domain, if there are any present for the parent domain.

Also, if `foo.com` does not have a value for one of these parameters, but the `.com` default distribution list settings do have values set, the `us.foo.com` default distribution list settings will reflect that, if it does not have values of its own.

If in the entire parent hierarchy of `us.foo.com`, this value is blank, Oracle WebMail will display a blank value.

A distribution list group type is set by the distribution list owner during list creation and controls distribution list attributes. Examples of distribution list attributes include what headers are included on mail delivered to a distribution list, or whether the list is moderated. The distribution list owner can change the group type after the list is created.

Table 2–1 describes the different group types.

Table 2–1 *Distribution List Group Types*

Type	Description
Announcement	Automatic replies or DSNs are not delivered to the sender. This group type is useful for making announcements to a group of recipients where the sender is not concerned with knowing if a message was delivered to individual recipients or if a recipient has set an automatic reply.
Discussion	Distribution list group type used for discussions among a group of people. Messages posted to this distribution list have a Reply-To header added, which is set to the distribution list mail ID itself. Thus, replying to such a message will send the reply back to the distribution list. Automatic replies and DSNs are delivered to the sender of a message (unless bounce processing is enabled for the list, in which case automatic replies or DSNs are sent to a special address <code>list_ID-bounce@domain</code> and are processed by the bounce processor in the List Server).
Edited	Distribution list group type that controls who can post mail to the list. Only users from a configurable list of editors can post mail to this list. Automatic replies and DSNs are delivered to the sender of a message (unless bounce processing is enabled for the list, in which case automatic replies or DSNs are sent to a special address <code>listid-bounce@domain</code> and are processed by the bounce processor in the List Server).

Table 2–1 (Cont.) Distribution List Group Types

Type	Description
Moderated	All messages posted to this distribution list group type are sent to list moderators (specified in the Group Moderators List attribute). At least one moderator must approve any message within 3 days for the message to be posted to the distribution list. Automatic replies and DSNs are delivered to the sender of a message (unless bounce processing is enabled for the list, in which case automatic replies or DSNs are sent to a special address <code>listid-bounce@domain</code> and are processed by the bounce processor in the List Server).

Subscription types control who can subscribe to a distribution list, as described in [Table 2–2](#).

Note: These subscription types do not apply to external distribution lists. For more information about external distribution lists, see ["External Distribution Lists"](#) on page 3-35.

Table 2–2 Distribution List Subscription Types

Type	Subscription
Open	No approval required; any user can subscribe.
Restricted	The list owner, or one member of a list of appointed approvers, must approve any request to subscribe to this distribution list.
Closed	Subscription requests are not processed. Users can only subscribe by invitation of the distribution list owner.

[Table 2–3](#) describes the distribution list posting type, which restricts whether nonmembers can post messages to the list.

Note: These posting types do not apply for edited lists.

Table 2–3 Distribution List Posting Types

Type	Description
Subscriber	Only list subscribers can post a message to the list. Mail from nonsubscribers is rejected.
Open	Both subscribers and nonsubscribers can post mail to the list.

Additionally, a server-side rule can be created to restrict certain users from posting messages to a particular list.

See Also: [Chapter 9, "Oracle Mail Server-Side Rules"](#) for more information about creating server-side rules

Distribution List Parameters

Distribution lists are created using the Oracle WebMail client. When a new list is created, various parameters are set, giving the new list various attributes, such as archiving and posting attributes.

[Table 2–4](#) describes distribution list parameters.

Table 2–4 Distribution List Parameters

Parameter	Description	Acceptable Values	Default Value
Distribution List Name¹	E-mail address of the new distribution list.	Any valid e-mail address	None
Owner¹	The e-mail address of the person who is to own the list.	A fully qualified e-mail address, such as <code>user@foo.com</code>	None
Maximum Message Size (bytes)	The maximum allowable size of a message posted to the distribution list. Messages exceeding this value are rejected. Use a value of 0 to indicate unlimited size.	Any integer greater than or equal to 0	None
Topic	Single-line phrase describing the topic of discussions on this list.	String of text	None
Invite Text	Multiline text sent in e-mail to users invited by a list owner to join the list.	String of text	None
Editor's List	List of users (e-mail addresses) for the editors of the list. Applicable only if the distribution list group type is Edited . See Also: Table 2–1, "Distribution List Group Types" for more details about group types	A fully qualified e-mail address, such as <code>user@foo.com</code>	None
Moderator's List	List of users (e-mail addresses) who are moderators of the list. Applicable only if the distribution list group type is Moderated . See Also: Table 2–1, "Distribution List Group Types" for more details about group types	A fully qualified e-mail address, such as <code>user@foo.com</code>	None
Group Approvers	List of users (e-mail addresses) who are the subscription approvers of the list. Applicable only if the distribution list subscription type is Restricted . See Also: Table 2–1, "Distribution List Group Types" for more details about group types	A fully qualified e-mail address, such as <code>user@foo.com</code>	None
Information Text	Multiline owner-provided descriptive text about the list	String of text	None
Merge Tag	A tag used for specifying mail merge and scheduler tags, enabling a list owner to support mail merge or scheduled mail delivery.		None
Collaboration Suite Database	If specified, the database on which the messages addressed to this list are queued until the List Server processes them. If not, then messages addressed to this list are queued wherever they are received.	Select an Oracle Database from the list	None
Auto Reconfirm	If set to True , requests to suspend or resume delivery from a distribution list trigger an e-mail from the list requiring a reply e-mail from the requester confirming the action. The default False value is recommended, because in the case of a suspend request, the requester may be unaware of the need to reconfirm the request and no longer be offline when the reconfirmation e-mail is delivered. In this case, suspension from the distribution list will not occur.	True or False	False

Table 2–4 (Cont.) Distribution List Parameters

Parameter	Description	Acceptable Values	Default Value
Group Type	Type of distribution list, as described in Table 2–1, "Distribution List Group Types" .	Discussion, Announcement, Edited, or Moderated	Discussion
Group Subscription Type	Type of subscription control placed on the list, as described in Table 2–2, "Distribution List Subscription Types" .	Open, Restricted, or Closed	Open
Post Type	Type of posting control placed on the list, as described in Table 2–3, "Distribution List Posting Types" .	Open or Subscriber	Open
Unsubscribe Not Allowed	If TRUE, allows only the list owner to unsubscribe a member from a list. If FALSE, anyone can unsubscribe.	True or False	False
List State	State of the list: active, inactive, or migrating: Active: Posting is permitted. Inactive: The list is not recognized as a recipient, no posting is permitted. Migrating: The list is being migrated into this installation.	Active, Inactive, or Migrating	Active
Enable Digests	If selected, enables the digest feature, in which posts to a list are held and only one e-mail message is sent to a list member at a member-specified frequency, containing all the posts to the list during the interim.	Not applicable (NA)	None
Enable Bounce Processing	If selected, when a DSN is received from any member of a list a specific number of times, the list owner receives a notification and can then unsubscribe the member from the list.	NA	None
Edit/View Member Privilege	Select from the list which type of user has the privilege to view and edit list members.	■ Owner ■ Owner, Approvers, Moderators, Editors ■ Members, Owner, Approvers, Moderators, Editors ■ All	Members, Owner, Approvers, Moderators, Editors
Restrict Membership to Group's Domain	If Yes, restricts distribution list membership only to those users within the domain of the distribution list. If No, users from outside the domain of the distribution list can be members.	Yes or No	Yes
Send Notification On Subscribe/Unsubscribe To	Indicates who receives notifications upon a successful subscription or unsubscription	None, Only Member, Only Owner, and Owner and Member	Owner and Member
Members from OiD Group	Select this option to tie the membership of the list to a group created in Oracle Internet Directory. All members of the Oracle Internet Directory group will be members of the new distribution list. Whenever members are added to the group in Oracle Internet Directory, those members will be automatically added to the distribution list.	NA	None

Table 2–4 (Cont.) Distribution List Parameters

Parameter	Description	Acceptable Values	Default Value
OiD Group	Select an Oracle Internet Directory group from the list. See Also: Chapter 9, "Dynamic and Static Groups in Oracle Internet Directory" in <i>Oracle Internet Directory Administrator's Guide</i> for more information about groups in Oracle Internet Directory	NA	None
NA	In the Suppress Headers in posts section, check the header types to be suppressed in mails distributed to the list.	NA	None
NA	In the Archive List Properties section, give the following information: ■ Group Has Archive: If set to True, enter an archive name in the Archive Name field in the form <code>ListArchive.list_name</code>. ■ If you choose to archive list information, select either Create a new archive on newsstore or Use existing archive. ■ Select from the Access Control list whether anyone can view archived messages, or restrict access to list members only	NA	False
NA	In the External List Properties section, select True from the Group is External list to make the list available externally. In the External Procedure field, enter a name for the external procedure used to resolve the list, in the following format: <code>schema-name.procedure-name</code> See Also: " External Distribution Lists " on page 3-35 for more information	NA	None

¹ Required information

See Also: "[Managing Distribution Lists](#)" on page 2-12 for more information

Creating a Distribution List

To create a list:

1. Open the Oracle WebMail client.

See Also: "[Oracle WebMail Client](#)" on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **List** subtab.
4. Click **Create a new list** in the **Distribution List Management** section.
5. Select the domain from the **Select Domain** list.
6. Select **Quick** or **List Server** from the **Distribution List Type** list. The distribution list type defines the distribution list type.
7. Click **Go**.

8. Enter the information in the corresponding fields, as described in [Table 2–4, "Distribution List Parameters"](#).
9. Click **Create**.

Modifying Distribution List Properties

To edit list properties:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **List** subtab.
4. Click **Edit/View List properties** in the **Distribution List Management** section.
5. Select the domain of the list from the **Select Domain** list.
6. In the **Search Criteria** field, enter the list name, or enter an asterisk (*) to display all available lists.
7. Click the name of the list you want to modify.
8. Edit the properties you want to change.
9. Click **Modify**.

Modifying Default Distribution List Parameter Values

An administrator can modify the default distribution list parameter values one sees when creating a new list. Modifying these parameter values is done at the domain level.

To modify default list parameter values:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **Domain** subtab.
4. Click **Default New List**.
5. Select the installation from the **Installation** list.
6. Select the domain from the **Domain** list and click **Submit** to display the Default New List Attributes page.
7. Enter the parameter values you want to set as defaults.

See Also: [Table 2–4, "Distribution List Parameters"](#) for a comprehensive list of distribution list parameters and descriptions for each

8. Click **Submit** to commit the changes, or use your browser's back button to return to the previous page.

Deleting a Distribution List

To delete a list:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **List** subtab.
4. Click **Delete list(s)** in the **Distribution List Management** section.
5. In the **Search Criteria** field, enter the list name, or enter an asterisk (*) to display all available lists.
6. Select the domain of the list from the **Select Domain** list.
7. Click **Go**.
8. Select the list you want to delete.
9. Click **Delete**.
10. Click **Return to Distribution List Management** to return to the Distribution List Management page.

Managing Distribution List Members

To add or delete users from a distribution list:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **List** subtab.
4. Click **Membership Management**.
5. Click **Add/Remove member(s)** to display the Choose a List to Manage Member page.
6. In the **Search Criteria** field, enter the list name, or enter an asterisk (*) to display all available lists.
7. Select the domain of the list from the **Select Domain** list.
8. Click **Go** to display the Add/Remove member(s) page.
9. Click the list in the **Name** column you want to manage.
10. Enter or remove fully qualified e-mail address or aliases in the following fields:
 - **Members (user):** Users on this system that are members of this list. Select users from the **User(s)** list and use the **Move**, **Move All**, **Remove**, and **Remove All** buttons to move users to and from the **Selected User(s)** list.
 - **Members (list):** Other lists that are members of this list
 - **Members (alias):** Aliases that are members of this list
 - **Members (foreign):** Users foreign to this system who are members of this list

11. Click **Modify** to apply changes.
12. Click **Cancel** to return to the Distribution List Management page.

Showing All Distribution Lists to Which a User Subscribes

To show all lists to which a user subscribes:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **List** subtab.
4. Click **Miscellaneous Functions**.
5. Click **Show all memberships of a user**.
6. Enter the user's name in the **User Name** field.
7. Select the domain of the user from the **Select Domain** list.
8. Click **Show Memberships** to display a list of distribution lists to which the user subscribes.
9. Click **Return to Distribution List Management** to return to the Distribution List Management page.

Managing Aliases

Aliases are shorter or more descriptive names you can use when sending messages to long user IDs or list names. Aliases are pointers that redirect messages to the intended receiver.

For example, if Jane Doe changes her name to Jane Roe, an [alias](#) can be created so that messages sent to her original account, `jane.doe@acme.com`, are automatically redirected to her new account, `jane.roe@acme.com`. This alias prevents her from losing messages sent to her old user ID.

Using the Oracle WebMail client, you can create, modify, and delete aliases.

This section includes the following topics:

- [Creating a New Alias](#)
- [Editing Alias Attributes](#)
- [Deleting Aliases](#)

Creating a New Alias

To create a new alias:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.

3. Click the **Alias** subtab to display the Alias Management page.
4. Click **Create New Alias**.
5. Select a domain from the **Select Domain** list.
6. Click **Go** to display the Create New Alias page.
7. Enter alias attributes in the following fields:
 - **Alias Name:** Enter a name by which the user ID will be referred. The alias name can either be just a name, such as dave, or it can be a fully qualified user or list ID, such as dave@foo.com.
 - **Alias Target:** Enter the fully qualified user or list ID to which the alias refers.
 - **Description:** Optionally, enter a description of the alias.
8. Click **Create** to display a confirmation page.
9. Click **Return to Alias Management**.

Editing Alias Attributes

To edit attributes of an existing alias:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **Alias** subtab to display the Alias Management page.
4. Click **Edit Alias Properties**.
5. Select the domain containing the alias from the **Select Domain** list.
6. Enter the alias name in the **Search Criteria** field. Enter an asterisk (*) to perform a wildcard search.
7. Click **Go**.
8. Click the alias you want to edit.
9. Edit the attributes, as necessary.
10. Click **Modify** to display a confirmation page.
11. Click **Return to Alias Management**.

Deleting Aliases

To delete an alias:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **Alias** subtab to display the Alias Management page.
4. Click **Delete Alias** to display the Choose Alias to Delete page.

5. Enter the alias name in the **Search Criteria** field. Enter an asterisk (*) to perform a wildcard search.
6. Select the domain containing the alias from the **Select Domain** list.
7. Click **Go**.
8. Select the alias or aliases you want to delete.
9. Click **Delete** to display a confirmation page.
10. Click **Return to Alias Management**.

Managing News Objects

If a **Network News Transport Protocol (NNTP)** server is configured on an Oracle Mail installation, an administrator can manage available peer servers and newsgroups using the Oracle WebMail client.

See Also:

- ["NNTP Server"](#) on page 3-41 for more information about configuring NNTP servers
- ["oesng"](#) on page D-6 and ["oespr"](#) on page D-9 for information about managing newsgroups and peers from the command line

This section includes the following topics:

- [Managing Peer Servers](#)
- [About Newsgroups](#)
- [Managing Newsgroups](#)

Managing Peer Servers

Use the Oracle WebMail client to add, edit, or delete **peer** servers, as the following sections describe:

See Also: ["About News Servers"](#) on page 3-42 for more information about peers

- [Adding Peer Servers](#)
- [Editing Peer Server Properties](#)
- [Deleting Peer Servers](#)

Adding Peer Servers

To add a peer server:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **News** subtab to display the Peer Server Management page.
4. Click **Add**.

5. Enter information in the following fields:
 - **Installation(s):** An Oracle Mail installation for which this peer is to be configured in which newsgroups have been created, and an NNTP server has been configured. The default is `um_system`.
 - **Hostname:** Fully qualified Internet host name of the peer, used by the NNTP inbound process to recognize an incoming connection as a peer connection
 - **Port:** NNTP port on which the peer listens, used by the NNTP outbound process to connect to the peer. The default is 119.
 - **Accepted Newsgroup(s):** Names of newsgroups (wildcards are acceptable) for which feed is accepted from this peer. If any groups are specified, messages are accepted only if addressed to one or more of them.
 - **Rejected Newsgroup(s):** Names of newsgroups (wildcards are acceptable) for which feed is rejected from this peer. If any groups are specified, messages are rejected if addressed to one or more of them. This is checked after **Accepted Newsgroup(s)**. If a newsgroup appears in both **Accepted Newsgroup(s)** and **Rejected Newsgroup(s)**, it is rejected.
 - **Feed Newsgroups:** Names of newsgroups for which feed is offered to this peer. If any groups are specified, then only messages posted to any of these groups are offered.
6. Click **Submit** to add the peer server and display a confirmation page. Click **Return to Peers** to return to the Peer Server Management page.
Click **Cancel** to return to the previous page.

Editing Peer Server Properties

To edit a peer server:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **News** subtab to display the Peer Server Management page.
4. Select an installation from the **Installation(s)** list.
5. Click **Go**.
6. Click the icon in the **Edit** column of the peer server for which you want to edit properties.
7. Edit the properties, as necessary.
8. Click **Submit** to commit the changes or **Cancel** to return to the previous page.

Deleting Peer Servers

To delete a peer server:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **News** subtab to display the Peer Server Management page.
4. Select an installation from the **Installation(s)** list.
5. Click **Go**.
6. Select the peer server you want to delete.
7. Click **Delete**.
8. Click **Yes** to delete the peer server or **No** to return to the Peer Server Management page.

About Newsgroups

Note: Newsgroups cannot be viewed in Oracle WebMail.

A **newsgroup** is a collection of messages discussing a particular subject, posted to an Internet site and redistributed through Usenet, a worldwide network of news discussion groups. There are two types of newsgroups, public and private:

- Public newsgroups have worldwide distribution, are maintained at many news sites, and are intended to contain nonsensitive information. The NNTP service exchanges newsgroup articles among news sites.

If your Oracle Mail installation hosts more than one domain, public newsgroups are shared, which greatly reduces the storage space required for news articles. An inbound server always services all public newsgroups configured for an Oracle Mail installation.

- Private newsgroups belong to and are visible in only one domain, where they are intended as internal discussion groups. Access to private newsgroups is permitted only if the inbound server has the local domain parameter set to that domain. An inbound server instance cannot host private newsgroups for more than one domain. For example, an inbound server instance with the local domain set to `acme.com` serves all public newsgroups in addition to only those private newsgroups that belong to the `acme.com` domain.

Newsgroups are organized into subject hierarchy. The first few letters of the newsgroup name indicates the major subject category; subcategories are represented by a subtopic name. Users can post to existing newsgroups and respond to previous posts, while only administrators can create newsgroups. Some newsgroups have a moderator, a designated person who decides which postings to allow or to remove.

Three attributes are associated with newsgroups:

- Name
- Description
- Posting allowed

Managing Newsgroups

Use the Oracle WebMail client to add, edit, or delete private and public newsgroups.

This section includes the following topics:

- [Adding Private Newsgroups](#)

- [Editing Private Newsgroup Properties](#)
- [Deleting Private Newsgroups](#)
- [Adding Public Newsgroups](#)
- [Editing Public Newsgroup Properties](#)
- [Deleting Public Newsgroups](#)

Adding Private Newsgroups

To add a private newsgroup:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Private Newsgroup Management**.
5. Select a domain on which the newsgroup will reside from the **Domain** list.
6. Click **Add** to display the Create/Edit Newsgroup page.
7. Enter information in the following fields:
 - **Newsgroup Name:** A string of text that is the name of the newsgroup
 - **Owner:** A fully qualified e-mail address of the person who is to be the newsgroup owner
 - **Description:** A string of text that briefly describes the newsgroup
 - **Moderated Newsgroup:** Select either *Yes* or *No* from the list depending upon whether the newsgroup is to be moderated. The default value is *No*.
 - **Moderator(s):** Fully qualified e-mail addresses of newsgroup moderators if the newsgroup is moderated
 - **Posting Allowed:** Select either *Yes* or *No* from the list depending upon whether posting to the newsgroup is allowed. The default value is *Yes*.
 - **Collaboration Suite Database:** Select from the list the database upon which news is stored for this newsgroup.
 - **Article Retention Day(s):** Enter a number to specify the numbers of days an article is to be stored on the database.
8. Click **Submit** to add the newsgroup and display a confirmation page. Click **Return to Private Newsgroups** to return to the Private Newsgroup Management page.

Click **Cancel** to return to the previous page.

Editing Private Newsgroup Properties

To edit private newsgroup properties:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Private Newsgroup Management**.
5. Select the domain on which the newsgroup resides from the **Domain** list.
6. To limit the number of newsgroups listed, enter a newsgroup name—or a partial name with an asterisk (*) as a wildcard—in the **Filter** field.
7. Click **Go**.
8. Click the icon in the **Edit** column of the newsgroup for which you want to edit properties.
9. Edit the properties, as necessary.
10. Click **Submit** to apply the changes and display a confirmation page. Click **Return to Private Newsgroups** to return to the Private Newsgroup Management page.
Click **Cancel** to return to the previous page.

Deleting Private Newsgroups

To delete a private newsgroup:

1. Open the Oracle WebMail client.

See Also: "[Oracle WebMail Client](#)" on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Private Newsgroup Management**.
5. Select the domain on which the newsgroup resides from the **Domain** list.
6. To limit the number of newsgroups listed, enter a newsgroup name—or a partial name with an asterisk (*) as a wildcard—in the **Filter** field.
7. Click **Go**.
8. Select the newsgroup you want to delete.
9. Click **Delete**.
10. Click **Yes** to delete the newsgroup or **No** to return to the Private Newsgroup Management page.

Adding Public Newsgroups

To add a public newsgroup:

1. Open the Oracle WebMail client.

See Also: "[Oracle WebMail Client](#)" on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Public Newsgroup Management**.

5. Click **Add** to display the Create/Edit Newsgroup page.
6. Enter information in the following fields:
 - **Newsgroup Name:** A string of text that is the name of the newsgroup
 - **Owner:** A fully qualified e-mail address of the person who is to be the newsgroup owner
 - **Description:** A string of text that briefly describes the newsgroup
 - **Moderated Newsgroup:** Select either *Yes* or *No* from the list depending upon whether the newsgroup is to be moderated
 - **Moderator(s):** Fully qualified e-mail addresses of newsgroup moderators if the newsgroup is moderated
 - **Posting Allowed:** Select either *Yes* or *No* from the list depending upon whether posting to the newsgroup is allowed
 - **Collaboration Suite Database:** Select from the list the database upon which news is stored for this newsgroup
 - **Article Retention Day(s):** Enter a number to specify the numbers of days an article is to be stored on the database
7. Click **Submit** to add the newsgroup and display a confirmation page. Click **Return to Public Newsgroups** to return to the Public Newsgroup Management page.
Click **Cancel** to return to the previous page.

Editing Public Newsgroup Properties

To edit a public newsgroup:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client
2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Public Newsgroup Management**.
5. To limit the number of newsgroups listed, enter a newsgroup name—or a partial name with an asterisk (*) as a wildcard—in the **Filter** field and click **Go**.
6. Click the icon in the **Edit** column of the newsgroup for which you want to edit properties.
7. Edit the properties, as necessary.
8. Click **Submit** to apply the changes and display a confirmation page. Click **Return to Public Newsgroups** to return to the Public Newsgroup Management page.
Click **Cancel** to return to the previous page.

Deleting Public Newsgroups

To delete a public newsgroup:

1. Open the Oracle WebMail client.

See Also: ["Oracle WebMail Client"](#) on page 1-2 for information about how to access the Oracle WebMail client

2. Click the **Administration** tab.
3. Click the **News** subtab.
4. Click **Public Newsgroup Management**.
5. To limit the number of newsgroups listed, enter a newsgroup name—or a partial name with an asterisk (*) as a wildcard—in the **Filter** field and click **Go**.
6. Select the newsgroup you want to delete.
7. Click **Delete**.
8. Click **Yes** to delete the newsgroup or **No** to return to the Public Newsgroup Management page.

Oracle Mail Servers

This chapter discusses the different servers of the Oracle Mail system.

A typical installation of Oracle Mail will have either Post Office Protocol (POP) or Internet Message Access Protocol (IMAP) servers running but not both. It is good practice to have two instances of each server running for availability. Note, however, an instance of a server consumes resources, such as memory, processing, and database connections. Not all of the processes must be running, either, as would be the case if an installation has only clients that access stored messages using IMAP servers. In this case, POP servers would be unnecessary and could be turned off and disabled.

In most cases, set all parameter values at the target level. Notable exceptions are the Housekeeper server and the SMTP Outbound server.

This chapter includes the following topics:

- [Oracle Database](#)
- [Managing Oracle Mail Servers and Instances](#)
- [Protocol Servers](#)
- [Housekeeper Server](#)
- [List Server](#)
- [NNTP Server](#)

Oracle Database

Oracle Mail uses the Oracle Database to store user folders and messages. The Oracle Database can be configured to store folders and messages for users in different domains. Messages destined for many accounts are stored only once, and links to the messages are set for all recipients. Single Oracle Databases can store mail for one domain or several different domains, while a large domain can be supported by multiple Oracle Databases.

Users can have one or more folders. These folders can be private, public, or shared. Private folders are visible only to the owner. Public folders are visible to all users in the owner's domain. Shared folders are visible to specified users and specified distribution lists.

Modifying Oracle Database Connection Parameters

The Oracle Database parameters are used by the Oracle Mail servers to connect to the Oracle Database.

To modify Oracle Database default parameters:

1. Open the Application Server Control.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Click a server in the **Name** column, such as **IMAP Server**, **POP Server**, or **List Server**.
5. In the **Target** section, click **Oracle Database Connection Parameters**.
6. Select the Oracle Database for which you want to make changes.
7. Modify the parameters listed in [Table 3-1](#) you want to change.

Table 3-1 Oracle Database Connection Parameters

Parameter	Option	Description
Timeout (seconds)	Enter a nonnegative number	Number of seconds before increasing the pool. The default value is 3 600.
Increment	Enter a nonnegative number	Number of database connections to be added to the connection pool. The default value is 1.
Minimum	Enter a nonnegative number	Minimum number of database connections in the connection pool. The default value is 1.
Maximum	Enter a nonnegative number	Maximum number of database connections in the connection pool. The default value is 10.
Alternate Connect String	String	Depending upon which server's database connection parameters are being modified, the alternate connect string is used by that server to connect to the Oracle Database chosen in Step 6. The databases shown on this page are the ones that are associated with this server or its instances on the Default Settings and Instance Settings pages. See Also: "Modifying Server Instance Default Parameter Settings" on page 3-7 or "Modifying Parameter Settings for a Specific Server Instance" on page 3-8 for information about associating specific Oracle Mail servers with Oracle Databases

8. Click **Apply**.

Managing Oracle Mail Servers and Instances

This section discusses how to start, stop, restart, refresh, and modify [servers](#) and [instances](#) using the Oracle Enterprise Manager Application Server Control Console.

Starting an Oracle Mail server starts all the instances constituting that server type, such as IMAP and POP.

Stopping an Oracle Mail server sends a command to the server instances to shut down. Whenever instances are created or certain parameter values are changed, the server must be stopped prior to any such action.

System maintenance, such as upgrading the server hardware or software, is another reason an administrator should stop a server. The Oracle Mail server instances cannot be running while this kind of upgrade is being performed.

Restarting an Oracle Mail server first stops the server and then restarts it. An administrator can restart a server after particular parameters have been modified in order to apply the changes.

Refreshing an Oracle Mail server notifies running instances to reread the configuration. Changed configuration values take effect but only for certain parameters, such as **Process Log Level**. Current instances do not stop. For example, the IMAP log level can be changed, the IMAP server refreshed, and the new log level will take effect without any current user sessions getting disconnected.

This section includes the following topics:

- [Log Files](#)
- [Enabling and Disabling Servers](#)
- [Starting, Stopping, Restarting, or Refreshing All Server Instances](#)
- [Creating a Server Instance](#)
- [Deleting a Server Instance](#)
- [Modifying Server Instance Default Parameter Settings](#)
- [Modifying Parameter Settings for a Specific Server Instance](#)

Log Files

Oracle Mail server logs files are located in the `$ORACLE_HOME/oes/log/server_type` directory. For example, List Server logs are found in the `$ORACLE_HOME/oes/log/list` directory. Each running server instance creates a log directory for itself when it starts and then writes to a log file in that directory.

For example, if a List Server instance is running on a UNIX platform, and the operating system process ID for the process is 12345, the log file for the process will be `$ORACLE_HOME/oes/log/list/12345/12345.log`.

The maximum size to which a log file can grow is configured by the **Maximum Log Size** parameter for a particular server. If a log file reaches the maximum size, it is renamed and a new log file is created. The old file is renamed by appending a number to the file name, such as `12345.log.000`.

The renaming process cascades to older files. The names of older files are changed by increasing the value of the appended number. [Example 3–1](#) shows a list of the log files in the log directory of the SMTP Outbound server.

Example 3–1 List of SMTP Outbound Server Log Files

```
% cd $ORACLE_HOME/oes/log/um_system/smtp_out/13067
% ls
13067.log      13067.log.002  13067.log.005  13067.log.008
13067.log.000  13067.log.003  13067.log.006  13067.log.009
13067.log.001  13067.log.004  13067.log.007
```

In the preceding example, log file renaming has occurred many times. The server instance is currently writing to `13067.log`, the next most recent log file is named `13067.log.000`, and the oldest log information is in the file named `13067.log.009`.

The number of old log files to keep is configured by setting the **Maximum Number of Log Files** server parameter. Once the maximum number of log files is reached due to log file renaming, the oldest file is deleted.

In [Example 3-1](#), if the **Maximum Number of Log Files** parameter is set to 10, and the size of the file named 13067.log grows to the maximum log size, the file named 13067.log.009 will be deleted, all remaining files will have their names changed, and a new 13067.log file will be created.

Log files for Oracle Mail servers are also found in the \$ORACLE_HOME/opmn/logs directory. These log files contain debug and error output from the servers. Usually, it is not necessary to look at these files, but they can be useful when you are investigating problems with servers.

[Example 3-2](#) shows where the administrator changes to the opmn log directory and lists the Oracle Mail server log files.

Example 3-2 Oracle Mail Server Log Files Listed in the opmn Log Directory

```
% cd $ORACLE_HOME/opmn/logs
% ls -l email*
bash-2.05$ ls -l email*
email~email_housekeeper~111669348732673252~1
email~email_imap~11166933915459529~1
email~email_imap~111669641725581757~1
email~email_listserver~111669349628087514~1
email~email_smtp_in~111669352320380482~1
email~email_smtp_in~111669739031461932~1
email~email_smtp_out~111669352925661397~1
email~email_smtp_out~111670781734121085~1
email~email_virus_scrubber~111669353532515101~1
```

[Example 3-3](#) shows the type of debug output available in these files.

Example 3-3 Debug Output

```
% tail -18 email~email_smtp_in~111669352320380482~1
-----
05/06/25 13:44:40 Start process
-----
successfully added instance 111669352320380482[0,1000,0,1000]
good; added service [ESSMI]
nslisten succeeded!
good; added registrar [(DESCRIPTION=(ADDRESS=(PROTOCOL=IPC) (KEY=OCS_
RGMUM300.appserver.acme.com)))]
good; mapped instance to all registrars
good; mapped service 0 to all registrars
complete addr is
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=127.0.0.1) (PORT=45963)) (PRESENTATION=ESS
MI) (SESSION=NS))
good; added handler [111669352320380482]
good; mapped handler 0 to all registrars
nsgrRegister succeeded!
INIT with 0 registrars.
SUCCESS with 0 registrars.
WAITEV with 0 registrars.
WAITEVP with 1 registrars.
FAILURE with 0 registrars.
```


Enabling and Disabling Servers

A server must be enabled before it can be started. Disabling a server prevents it from being started.

To enable or disable a server:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select a server to enable or disable.
5. Click **Enable** or **Disable**.

Note: Disabling a server that is currently running will first stop the server, then disable it.

See Also:

- ["Enabling an Oracle Mail Server"](#) on page D-19 for information about enabling a server from the command line
- ["Disabling an Oracle Mail Server"](#) on page D-20 for information about disabling servers from the command line

Starting, Stopping, Restarting, or Refreshing All Server Instances

To start, stop, restart, or refresh all instances of a particular server:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select a server to start, stop, or refresh.
5. Click **Start**, **Stop**, **Restart**, or **Refresh**.

Click **Restart** to apply any changes made to any server instance parameters.

Restart shuts down all instances of the selected server before starting them up again. This will result in a brief interruption of e-mail service.

Click **Refresh** to apply changes made to log levels or debug flags. **Refresh** does not shut down instances of the server and will not interrupt any service.

Note: The refresh icon located on this page refreshes the information on this page, not any servers.

See Also: ["Starting All Instances of an Oracle Mail Server"](#) on page D-21 for information about starting all server instances from the command line

Creating a Server Instance

To create a server instance:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select the server for which the new instance is to be created.
5. Click **Stop** to stop the server.
6. Click the server in the **Name** column to display the page for that server.
7. Click **Create**. This creates a new server instance that inherits the default parameters.
Optionally, click **Create Like** to create a new server instance with the same parameters as an existing, previously selected server instance.
8. Click the newly created instance to modify the parameter values.
9. Click **Apply** when finished.
10. Click **Clear Instance Settings** to clear parameter settings for all process instances listed.
11. Return to the Mail Application page.
12. Select the server for which the new instance was created.
13. Click **Start** to bring up the server.

See Also:

- ["Modifying Server Instance Default Parameter Settings"](#) on page 3-7 for more information
- [Appendix F, "Oracle Mail Server Parameters and Log Files"](#) for descriptions of the various server instance parameters

Deleting a Server Instance

Caution: Deleting an Oracle Mail server instance may disable some or all e-mail processes.

To delete a server instance:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select the server from which the instance is to be deleted.
5. Click **Stop** to stop the server.
6. Click the server in the **Name** column to display the page for that server.
7. Select the server instance you want to delete.
8. Click **Delete**.

Modifying Server Instance Default Parameter Settings

All new server instances that are created inherit the default parameter values for that server type, unless the instance is created using the **Create Like** button. The parameters, both default and replicated, can later be modified for specific server instances.

Notes:

- In most cases, administrators should modify parameters only at the target level to reduce the chances of creating conflicts between two instances of the same server. Two notable exceptions to this are the Housekeeper server and the SMTP Outbound server.
 - Servers must be restarted or refreshed whenever parameters are modified.
-
-

See Also:

- ["Creating a Server Instance"](#) on page 3-6
- [Appendix F, "Oracle Mail Server Parameters and Log Files"](#) for descriptions of the various server instance parameters

To modify default server instance parameter settings:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.

4. Click a server in the **Name** column, such as **IMAP Server**, **POP Server**, or **List Server** to display the page for that server.
5. Click **Default Settings** in the **Target** section.
6. Modify the parameters you want to change or click **Revert** to reset the parameters to their default values.
Click **Cancel** to cancel any changes and return you to the server page.
7. Click **Apply** to apply changes.
8. Restart the server to apply any changes prior to creating any new server instances.

See Also: ["Starting, Stopping, Restarting, or Refreshing All Server Instances"](#) on page 3-5

Modifying Parameter Settings for a Specific Server Instance

Notes:

- Servers must be restarted whenever instance parameters are modified in order for the changes to be applied.
 - Whenever a setting is cleared at the instance level, the corresponding value from the default parameter settings page is used. This inheritance occurs for all Oracle Mail servers and Oracle WebMail.
-
-

To modify parameters for a specific server instance:

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Click a server in the **Name** column, such as **IMAP Server**, **POP Server**, or **List Server** to display the page for that server.
5. Click the server instance you want to modify.
6. Modify the parameters you want to change or click **Revert** to reset the parameters to their default values.
Click **Cancel** to cancel any changes and return you back to the server page.
7. Click **Apply** to apply changes.
8. Restart the server to apply any changes.

See Also: ["Starting, Stopping, Restarting, or Refreshing All Server Instances"](#) on page 3-5

Protocol Servers

The scalable protocol server programming framework maintains a pool of worker threads that handle the work for the clients. In addition, a pool of database connections are shared across client connections. Incoming client requests have worker threads assigned to them. Worker threads read client commands, obtain database connections, and carry out operations. Once the database connections are released back to the pool, the threads return to the worker thread pool.

A system can contain multiple mail stores, and the IMAP and POP servers can be configured to create database connection pools to more than one mail store. Administrators use the IMAP and POP server parameters to control connection pool sizes.

This section includes the following topics:

- [IMAP and POP Servers](#)
- [SMTP Server](#)

IMAP and POP Servers

The **Internet Message Access Protocol (IMAP)** (the current version is IMAP4) provides functionality to manipulate mail messages and mail folders stored on the server and to enable an offline client to resynchronize with the server. IMAP also has primitives enabling optimization of online performance, especially for large MIME messages.

The **Post Office Protocol (POP)** (the current version is POP3) provides mail manipulation services for smaller Internet nodes where it can be impractical to maintain a message transport system or undesirable to keep an Internet connection open for long periods of time. Messages are temporarily stored on the server until they are downloaded to a client system.

Note: Either the IMAP server or the POP server will be enabled in a given installation. Disable the unused server.

This section includes the following topics:

- [IMAP and POP Server Architecture](#)
- [IMAP and POP Session Data Flow](#)
- [Oracle WebMail Client Session Data Flow](#)
- [IMAP and POP Server Instance Parameters](#)
- [Managing IMAP and POP Servers](#)

IMAP and POP Server Architecture

The IMAP and POP servers obtain the benefits of multithreading, database connection sharing, and load balancing by using the framework. These benefits enable the servers to support thousands of concurrent user connections while using very few system resources.

Many operating systems limit the number of file descriptors and sockets that a single instance can open. These limits can make it necessary to run more than one instance of an IMAP or POP server, in which case the listener distributes the load between them.

Administrators must verify the correctness of the operating system parameter controlling the file descriptors for such instances.

See Also: *Oracle Collaboration Suite Installation Guide for Solaris Operating System* for more information about operating system parameters

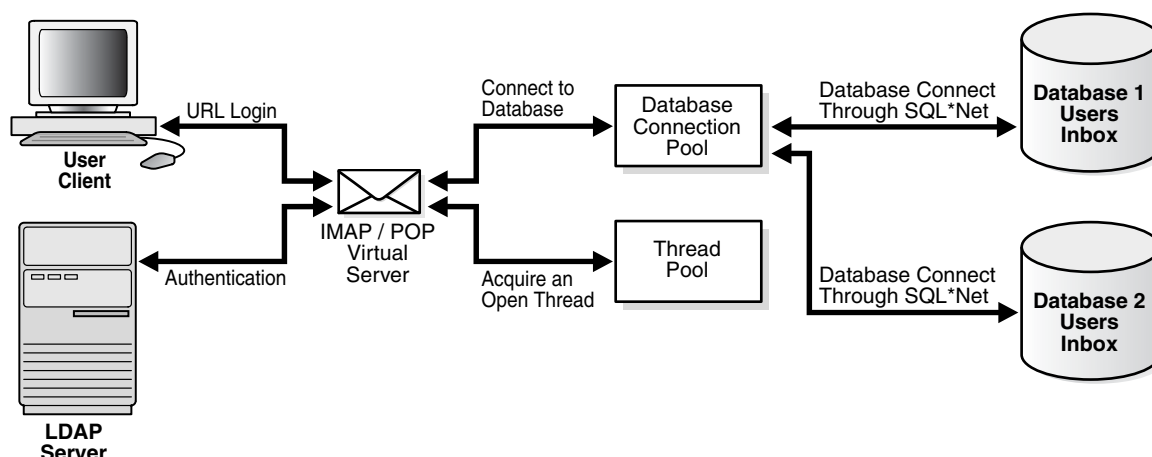
IMAP and POP Session Data Flow

As a user connects, either through various clients, such as Microsoft Outlook or Mozilla Thunderbird, or the Oracle WebMail client, it is important to understand the connection flow and the components the clients are accessing during login and during an individual session.

All IMAP-compliant and POP-compliant clients are able to access the Oracle Mail server. These clients must be directly installed on the end-user client's system and configured by that user.

The access method flow is described, as follows, and shown in [Figure 3-1](#):

1. The client starts an application, such as Microsoft Outlook or Mozilla Thunderbird. The client must be configured to connect to the system where IMAP and POP are listening.
2. The user is prompted to enter a user name and password.
3. Authentication of the login information is checked against the Oracle Collaboration Suite Infrastructure through the IMAP server connection for validation. A connection is opened from the LDAP connection pool of the IMAP server. Once authentication is validated, the LDAP connection is released back to the original connection pool.
4. The user IMAP connection is passed to the database connection pool for access to the user e-mail account.
5. Once a connection has been established, a download of the user's INBOX begins. Any new incoming messages are also posted to the INBOX for sorting.
6. When the INBOX download is complete, the client releases the database connection back to the database connection pool, but maintains an authenticated cached connection to the IMAP instance.
7. Now when there is a call to the database, such as when a user opens a message or opens a folder, a new connection to the database is established to respond to the request. Once the request is satisfied, the database connection is released.

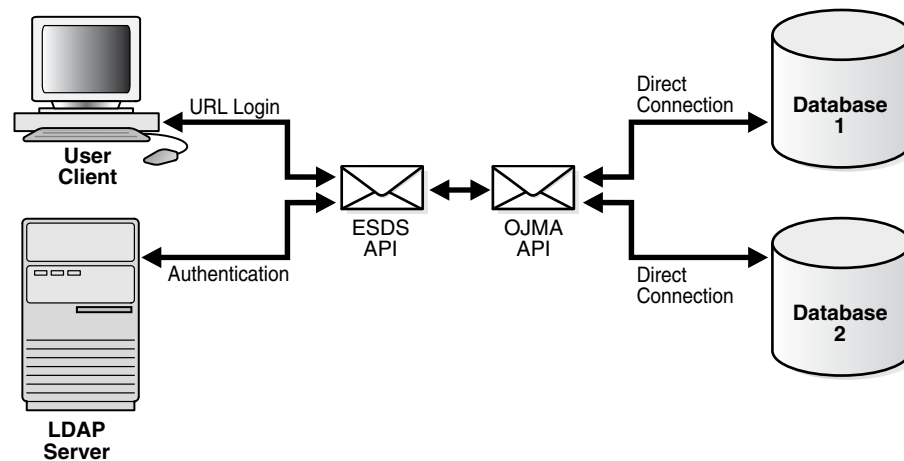
Figure 3–1 IMAP and POP Data Flow

Oracle WebMail Client Session Data Flow

The Oracle WebMail client runs on the Oracle Collaboration Suite Applications tier. It is not necessary for the user to download any information, except for a cookie to the local client. Besides entering a URL in a Web browser, no local configuration is necessary. However, the authentication method is similar to that of other clients, except that the Oracle WebMail client caches the user preferences. The access method is through Oracle Java Mail API (OJMA) and is not, by default, using the IMAP server. IMAP can be configured to use IMAP through the `oc4j.properties` file under the `$ORACLE_HOME/j2ee/OC4J_OCSCClient/config` directory.

The access method flow is described, as follows, and shown in [Figure 3–2](#):

1. The client starts a browser and enters a given URL. This URL connects to the Applications tier where the Oracle WebMail client is configured.
2. The user is prompted to enter a user name and password. Once committed, the client calls the ESDS application programming interface (API).
3. The ESDS API uses the authentication information to check against the Oracle Collaboration Suite Infrastructure for validation. The connection is opened through the ESDS LDAP connection pool. Once authentication is validated, the LDAP connection is closed.
4. Once authenticated, the connection is passed to the OJMA connection pool for direct access to the user e-mail account.
5. Once a connection has been established, a download of the user's INBOX begins. Any new incoming messages are also posted to the INBOX for sorting.
6. When the INBOX download is complete, the database connection is released back to the database connection pool, but maintained as an authenticated user by OJMA.
7. Now when there is a call to the database, such as when a user opens a message or opens a folder, a connection from the pool is opened by OJMA and the request is satisfied. Once the request is satisfied, the database connection is released.

Figure 3–2 Oracle WebMail Client Data Flow

IMAP and POP Server Instance Parameters

See Also: ["Oracle Mail IMAP Server"](#) on page F-6 and ["Oracle Mail POP Server"](#) on page F-18 for detailed information about IMAP and POP server instance parameters

Managing IMAP and POP Servers

See Also: ["Managing Oracle Mail Servers and Instances"](#) on page 3-2 for instructions on creating, deleting, and modifying server instances

SMTP Server

The Simple Mail Transfer Protocol (**SMTP**) enables e-mail messages to be sent between servers, and is used by most Internet e-mail systems. Mail clients use SMTP to send messages to a mail server, and use either POP or IMAP to retrieve messages.

The SMTP server handles all inbound and outbound mail, implements the SMTP protocol, and interacts with the **domain name server (DNS)** and the Oracle Internet Directory server for information about hosts and users.

The SMTP server is also responsible for delivering e-mail messages to archive servers based on configured archive policies.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about archive policies

This section includes the following topics:

- [Various SMTP Configurations](#)
- [SMTP Message Flow](#)
- [SMTP Inbound Server Architecture](#)
- [SMTP Outbound Server Architecture](#)
- [SMTP Address Rewriting Rules](#)

- [SMTP Server Parameters](#)
- [Managing SMTP Servers](#)
- [SMTP Routing Control](#)

Various SMTP Configurations

The flexible architecture of Oracle Mail enables users to set up a single or multitier configuration appropriate for specific a site.

This section includes the following topics:

- [Single Node Setup](#)
- [Single Oracle Database Setup](#)
- [Multiple Oracle Database Setup](#)

Single Node Setup A single node setup has one Oracle Database and one SMTP server running on the same host, supporting a small numbers of users.

Single Oracle Database Setup A single Oracle Database setup divides two servers into two tiers:

- The tier where the Oracle Database resides
- The Applications tier where SMTP and other protocol servers reside

This configuration provides fault tolerance and the flexibility to run multiple SMTP servers with distributed loads by running the servers behind a network director. Alternatively, it can have multiple **MX records** for the domain.

Multiple Oracle Database Setup A multiple Oracle Database is a scaled version of a single Oracle Database setup that supports a large user base. The multiple database setup consists of a set of hosts, each running an Oracle Database, and the Applications tier, which includes a set of hosts running the SMTP server and other protocol servers. Each SMTP inbound server can accept and deliver mail to multiple databases. The SMTP outbound server performs queue processing against a single database and delivers to multiple databases. It is possible to have one database just to receive the mail and the rest supporting the user base for added fault tolerance and reduced contention.

Each SMTP server serves only one Oracle Database, and each Oracle Database must have an SMTP server. The Oracle Databases on the SMTP hosts are used as SMTP queues and do not contain users.

SMTP Message Flow

The SMTP inbound server is responsible for handling the incoming SMTP connection. It receives incoming messages, queries the Oracle Internet Directory server to find and authenticate the addresses, rewrites addresses based on the rewriting rules, and applies antispam rules. If all the steps are successful, the SMTP **message transfer agent (MTA)** accepts the message and inserts it into the corresponding queue based on the destination address, as shown in [Figure 3-3](#).

If the message recipient is a user outside of the system, the message is stored in the relay queue to await further processing. If the recipient is local, the message is stored in the local delivery queue. The **Local Domains** parameter contains the list of local domains used to determine whether an address is local. The local delivery module picks up the message later, applies the rules, and delivers it to the user's inbox.

If administrators do not want to process the messages immediately, they can be stored in the submission queue and marked as submitted or unprocessed. Messages created by the software developer's kit (SDK) applications are also placed in the submission queue and marked as submitted. Messages in the submission queue are picked up by the SMTP outbound server.

For relay messages, the SMTP outbound server queries the DNS server, applies any rules to the messages, and sends them out using SMTP. For submitted messages, processing by the address rewriting and DNS resolution module happens first. Subsequently, the SMTP outbound server sends them to the local delivery queue or to the Internet, depending on whether the messages recipients are local.

During address resolution, the server determines whether the message is addressed to a [distribution list](#) handled by the List Server. If so, the server places the message in the queue for the [List Server](#), which then picks up the message, expands the distribution list, and delivers the message.

Messages for users on a different Oracle Database are placed in the relay queue. The outbound server picks up and delivers the messages to the SMTP inbound server for the other Oracle Database.

If the recipient is determined to be local, the message is stored in the local delivery queue. To determine if an address is local, the **Local Domains** parameter is used. This parameter contains the list of domains that are considered local.

If the recipient is an outside user, either on another Oracle Database or on the Internet, the message is stored in the relay queue to await further processing by the SMTP outbound server.

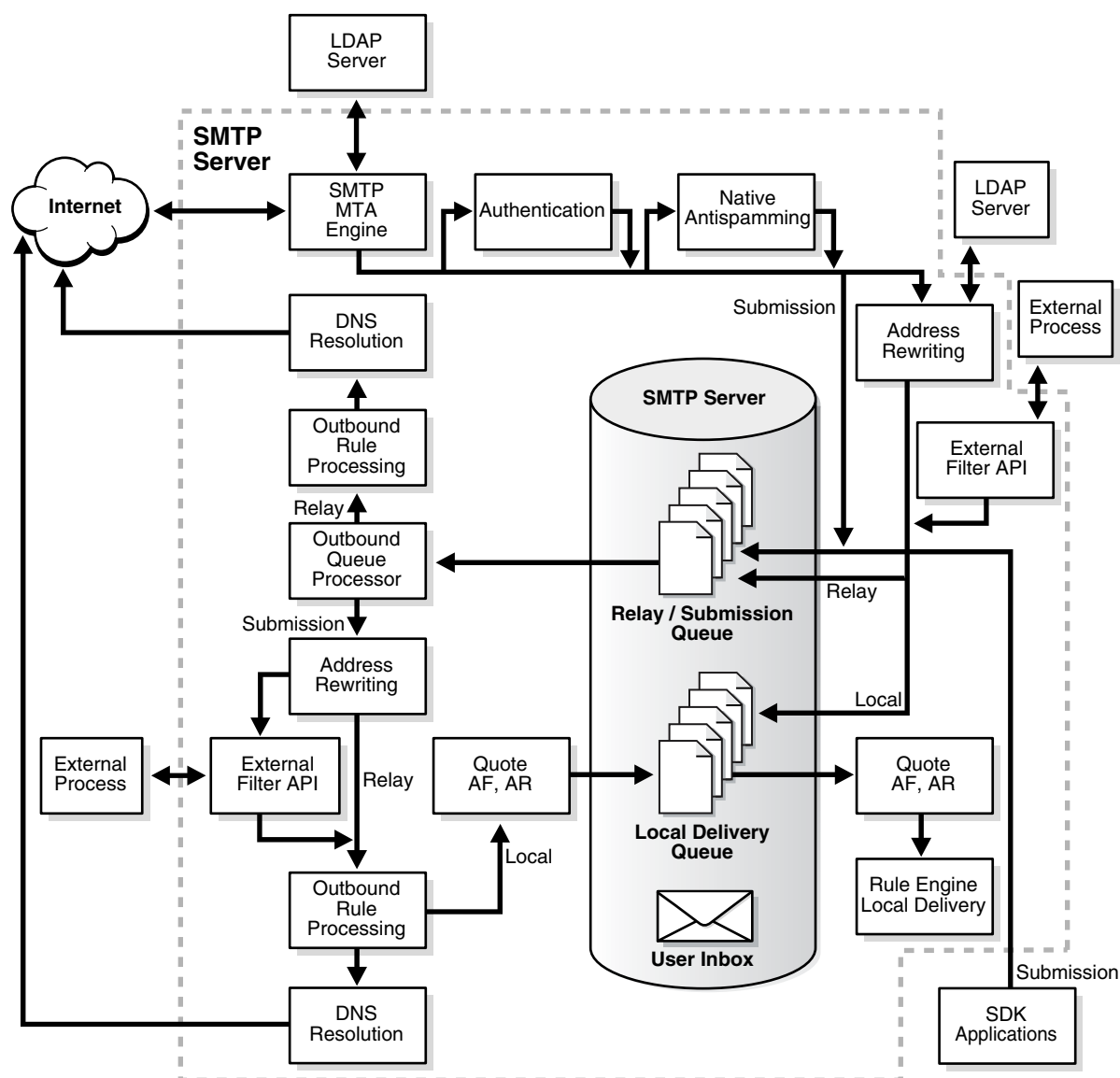
See Also: ["SMTP Address Rewriting Rules"](#) on page 3-17 and ["List Server"](#) on page 3-33 for more information

The SMTP server is also responsible for delivering e-mail messages to archive servers based on configured archive policies. To enable archive delivery, set the **Archive Processing** parameter to `Enabled` in both SMTP inbound and outbound servers and the List Server. This parameter determines whether the servers will check all messages for archive eligibility.

Note: The **Archive Processing** parameter must contain the same value for all the Applications tiers to ensure that archive message processing is consistent. In addition to the **Archive Processing parameter**, also set **Archive Queue Processing** to `Enabled` for the SMTP outbound server. This parameter allows the SMTP Outbound server to generate and deliver archive messages.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about enabling the archive feature

Figure 3-3 SMTP Message Flow



SMTP Inbound Server Architecture

The SMTP inbound server listens for client requests, processes incoming messages, and either delivers them locally or places them into queues for further processing.

The Oracle Net listener listens for incoming client requests on the SMTP port (default 25) and transfers connections to the SMTP server. The SMTP server maintains three thread pools to perform its tasks:

- Worker thread pool, through which it handles client requests
- Oracle Internet Directory server connection pool, through which it performs user authentication and address resolution
- Database connection pools to the configured Oracle Databases, through which it delivers local messages

Note: If the recipient is an SMTP distribution list, the message is placed in the submission queue to be processed. The SMTP outbound server expands SMTP distribution lists and performs delivery to the expanded lists.

Upon receipt of a connection request, the SMTP server picks up a worker thread from the pool to handle the request and:

- It performs name resolution on the incoming message using a connection from the Oracle Internet Directory.
- If the SMTP server is enabled for archiving, it determines whether archiving is needed for sender or recipient addresses. If message archiving is enabled, the message is placed in the archive queue for further processing.

See Also: ["Oracle Mail Archive Policies"](#) on page 8-26 for more information about enabling the archive feature

- It applies antivirus and antispam rules.

See Also: ["Oracle Mail Routing Control"](#) on page 8-8 for more information about these rewriting rules

- It applies recipient rewriting rules to the e-mail addresses to determine whether the message is to be delivered locally, sent to another Oracle Database, or sent out to the Internet.
- The SMTP MTA accepts the message and inserts it into the corresponding queue based on the destination address.
- It terminates the SMTP connection to the client for all but the local recipients and the worker thread continues to process the message and perform local delivery.

SMTP Outbound Server Architecture

The SMTP outbound queue processor processes messages in the submission, local, and relay queues. It has a main thread for each queue that periodically polls the database for messages in its queue. Whenever there are messages to process, a new thread is spawned to process the mail.

The outbound queue processor also maintains two other thread pools:

- The Oracle Internet Directory server connection pool, through which the outbound queue processor performs address authentication. If `SMTP_auth` is turned on, this thread pool is also used for user authentication prior to sending out a message.
- The database connection pool to the mail store, through which the outbound queue processor delivers local messages.

The size of all of these thread pools can be set through the Application Server Control Console.

See Also: ["Modifying Parameter Settings for a Specific Server Instance"](#) on page 3-8 for more information about setting server parameters

When a thread is started to process a message from one of the queues, it picks up a database connection from the pool and gets connections from the Oracle Internet

Directory pool as needed. After the mail is processed, the database and Oracle Internet Directory threads are returned to the pool.

Messages in the submission queue are treated as not yet processed, and so must go through the antivirus and antispam rules and rewriting rules to determine their destination. After processing, the messages are either placed in the local delivery queue or in the relay queue.

If the recipient is a user outside of Oracle Mail, the message is stored in the relay queue to await further processing by the SMTP outbound server. If the recipient is on an Oracle Database different from the database that was used to insert the incoming message, and if the SMTP inbound server is configured to perform delivery to this database, the message is copied and delivered into this database using SQL*Net. If the SMTP inbound server is not configured to deliver to the recipient's Oracle Database, the message is placed in the relay queue to await further processing by the SMTP outbound server.

Messages in the local delivery queue are destined for a local mailbox, so the queue processes or applies local rewriting and other rules, if any, and inserts the mail into the user's inbox in the database using a connection from the database pool.

Relay messages require further processing because their recipients are either on a different local message store or outside the e-mail system. Relay messages first go through the sender rewriting rules, then the system rules are invoked for event relay and external filter processing. If the system rule processing and external filter processing are successful, the DNS resolution takes place. The SMTP outbound queue processor then sends the relay messages to another Oracle Database in the system or to the Internet using SMTP, depending on whether the message recipients are local or not.

If the delivery of an e-mail fails, the message is returned into the queue and delivery is retried after intervals defined by the `minqueueage` parameter. If the attempted redeliveries are unsuccessful during the interval equal to the `queuetimeout` parameter, a delivery failure message is sent to the sender.

SMTP Address Rewriting Rules

The SMTP address rewriting rules enable you to check and correct the addresses of an e-mail message before sending it to its final recipient destinations. Rules resolve a focused or internal format address into a mailer-host-user triplet that can be delivered.

[Table 3–2](#) lists and describes mailer, host, and user parameters.

Table 3–2 Mailer, Host, and User Parameters

Parameter	Description
Mailer	Specifies the Oracle Mail SMTP server daemons used for delivery. Note: This is the only mailer available. Creating alternative mailers is not a feature of this application.
Host	Specifies either a fully qualified host name, such as <code>hostname.acme.com</code> , or a domain name, such as <code>foo.com</code> .
User	Specifies the recipient user name.

To execute and complete name resolution, the SMTP inbound server parses each rule for every address in the message envelope during mail delivery.

This section includes the following topics:

- [Components of Rewriting Rules](#)

- [Rule Execution Guidelines](#)
- [Oracle Mail Rewriting Rules](#)
- [Creating an Oracle Mail Address Rewriting Rule](#)

Components of Rewriting Rules Headers for the message and the envelope are distinctly different. The envelope headers are generated by the receiving e-mail application, rather than by the sender. Received: headers are the envelope headers, and relate only to the envelope From and envelope To fields.

The envelope From header is created from the MAIL FROM entry in the received message. For example, when a sending computer puts MAIL FROM: jsmith@foo.com in the message, the receiving e-mail application generates the envelope From field content as jsmith@foo.com.

Similarly, the envelope To is derived from an incoming message line, such as RCPT TO: john.smith@foo.com. The information for envelope To and envelope From is stored in a different location from the header.

Mail is routed based solely on the envelope To data rather than on the message To: or From: headers supplied by the sender. These headers contain no significant envelope information and can misrepresent who sent the mail to whom. An example of a message header as is illustrated in [Figure 3-4](#).

Figure 3-4 Message Header

```
From: john.doe@foo.com                      Fri 4:17 PM
To: fred.jones@uuhost
Subject: Re: New Message
CC:
```

A handshake between two SMTP systems executing transactions through port 25 involves a series of action dialogs for each message being delivered. These messages can be seen on the receiving or sending systems only by running in debug mode, as illustrated in [Example 3-4](#). The example illustrates why the routing of mail ignores the Message From and Message To headers, which can be faked, and relies instead on the envelope data.

Example 3-4 Example of Original Header

```
HELO acme.org
250 mail.rico.net Hello ernie.com [104.65.21.123], pleased to meet you
MAIL FROM: forged-address@foo.org
250 forged-address@foo.org... Sender ok
RCPT TO: john.smith@foo.com
250 john.smith@foo.com... Recipient OK
DATA
354 Enter mail, end with "." on a line by itself
From: another-forged-address@moreover.com
To: (your address suppressed for stealth mailing and annoyance)
.
250 OAA08757 Message accepted for delivery
```

[Example 3-5](#) shows how the header illustrated in [Example 3-4](#) is seen by the message recipient.

Example 3–5 Resulting Header as Seen by the Recipient

```
Received: from foo.org ([104.128.23.115]) by mail.rico.net (8.8.5)
From: another-forged-address@moreover.org
To: (your address suppressed for stealth mailing and annoyance)
```

Notice that the only true data seen by this recipient is in the `Received` line, which was taken from the `RCPT TO` entry actually sent. The apparent sending addresses need not have any relationship to the physical facts. They are taken from the data of the message `MAIL FROM` and `RCPT TO`: lines exactly as entered by the sender, with no necessary relationship to what is factual.

The preceding examples illustrate why the `MAIL FROM`: and `RCPT TO`: headers are not reliable in mail, because they can easily be forged.

Rule Execution Guidelines Address rewriting rules are applied sequentially, beginning with rule 1. All rules are applied, unless a result starts with a dollar sign (\$) and at sign (@) combination (\$@), which immediately stops rule execution and ignores any remaining rules. If a rule has a syntax error or cannot be executed, it is ignored.

A rule is applied to its own output in a loop until the application of the rule does not yield anymore changes in the result. The next rule in the sequence is then applied. After all the rules have been executed, an Oracle Internet Directory resolution is performed on the result. If the Oracle Internet Directory resolution returns a changed address due to an alias (for example, the address rewriting rules are applied to the changed address), the Oracle Internet Directory resolution is performed again. When the Oracle Internet Directory resolution rule does not yield any more changes, the rule execution process is done.

Oracle Mail Rewriting Rules To understand rewriting rules, you must understand their components: a left-hand side (LHS), a right-hand side (RHS), and a description used in this format:

```
Pattern (LHS),Result (RHS), Description
```

[Table 3–3](#) lists descriptions of the rewriting rule components.

Table 3–3 Rewriting Rule Components

Format	Description
Pattern (LHS)	Specifies the pattern to be changed.
Result (RHS)	Specifies that whenever the Pattern (LHS) is seen, it is changed with this result.
Description	Specifies the administrator's rule notation, and is not used in address name resolution. Anything after the last comma does not require quotation marks.
Comma (,)	Separates the LHS, RHS, and Description. No spaces are allowed between the commas, nor before the first comma.

When the `Pattern (LHS)` is compared against the address and finds a match, the `Result (RHS)` replaces that match in the address. The comparison is not case-sensitive. If no match is found, then this rule is skipped and the next rule is applied. A rule can be applied to an address resulting from applying a previous rule.

Tokens and Matching

When processing an address for rewriting by a rule, the SMTP daemon first separates the address into parts called tokens and stores them into a buffer called the workspace.

A rule's **Pattern (LHS)** is also divided into tokens, which are then compared to the tokens in the workspace. If the two sets of tokens are identical, it is a match, and the result of the left-hand side comparison is true.

LHS Operator Descriptions

If rules always had to match addresses exactly, too many rules would be required, which would render their usage unproductive. Instead, operators such as wildcards can also be used to match arbitrary text in the workspace. To make the entire **Pattern (LHS)** match, wildcard operators match as little as possible. [Figure 3-5](#) shows an example of rewriting rule.

Figure 3-5 Rewriting Rule Example

LHS	RHS	Description
\$*.\$*	\$1.\$2@uuhost,	Changing the first.last to first.last@uuhost

The following LHS operators are used as wildcards or token identifiers:

- **\$*** = zero or more tokens, and prefers zero, or the fewest possible, to satisfy a **Pattern (LHS)** match

For example:

```
fred.jones resolved by the Pattern (LHS) rule
Result (RHS) rule = fred.jones@uuhost
```

- **\$+** = one or more tokens, and prefers one, or the fewest possible, to satisfy a **Pattern (LHS)** match

To illustrate, consider passing the address `john.jones@home.ORG` to the **Pattern (LHS)**, as shown in [Figure 3-6](#):

Figure 3-6 Changing Uppercase to Lowercase

LHS	RHS	Description
\$* @ \$+ .ORG,	\$1@ \$2.org,	Changing the uppercase to lower case

For example:

```
john.jones@home.ORG resolved by the Pattern (LHS) rule,
Result (RHS)= john.jones@home.org
$* (matches zero or more) = john.jones
@ matches exactly
$+ (matches one or more) = home
```

- **\$-** = exactly one token

`john` rewritten by the right-hand side token `rhs $- = ->john`

If the result was `john@uuhost`, then the rule would not match.

RHS Operator Descriptions

The following list describes the various RHS operators:

- **\$1, \$2** identifies **Pattern (LHS)** tokens to be passed over into the **Result (RHS)**. These are copied by position from the **Result (RHS)** location.

- lhs `$*.$*` where rhs `$1.$2`
- `$:` Indicates that the rule should be applied only once.
- rhs `john` rewritten by `$:$1 = john`
- `$@:` Exactly none. Rules are not applied beyond this point if the `$@` operator is reached during the rewriting rule processing.
- `$!:` Reject: A DSN will be sent if the SMTP outbound server is rewriting
- `$%:` Discard: The address is discarded. No DSN is sent if the SMTP outbound server is rewriting.

Creating an Oracle Mail Address Rewriting Rule Oracle Mail uses two types of address rewriting rules:

- Sender rewriting rules: Apply only to senders of messages that are relayed out by the SMTP outbound server
- Recipient rewriting rules: Apply to all incoming and outgoing recipient addresses

Rules can be written using the Application Server Control Console and are executed in the order in which they are entered.

Examples of Rewriting Rules

The following example takes the `From:` (sender) and the `To:` (recipient) addresses and rewrites them using the rewriting rules.

- Sender Rewriting Rules

```
$*@$+.com,$1@uuhost.com, "This changes john.doe@foo.com to john.doe@uuhost.com"
```

Rule:

1. Match anything before the at sign (@) and take the one token after the at sign (@) with the `.com` at the end and change it.
2. Keep the user name and pass it to the RHS through `$1`, which is in direct order or the first token from the LHS, `john.doe` and pass the @ sign as is, but change the `$2` token (second token) to `uuhost.com`.

The receiving SMTP daemon accepts this message, and accepts `john.doe@uuhost.com` as the sender of the message. It is important to remember that the header information is never changed from its original entries.

- Recipient Rewriting Rules

```
$*.$*@uuhost,$1.$2@foo.com, "This changes fred.jones@uuhost to fred.jones@foo.com"
```

Rule:

1. Capture both the first name and the last name of any address that has `uuhost` after the at sign (@).
2. Bring those tokens over as `$1` and `$2` respectively, and keep a period (.) between them.
3. After the at sign (@), replace `uuhost` with `foo.com`.

- Rewriting an E-mail Address

The following example shows how to rewrite an e-mail address using fred.jones@uuhost. The address points to uuhost, which is a UNIX to UNIX copy (UUCP) system name. The message is sent using the UUCP software which requires the address form of uuhost!username and that the current address be rewritten for UUCP. Consider the following example:

```
$*@uuhost,uuhost!$1,"Changing from to UUCP address"
```

Using the rewriting rule shown in [Figure 3-7](#), you can change this address to a more compatible Internet address, such as fred.jones@foo.com.

Figure 3-7 Message Flow through Rewriting Rules

LHS	(S)	RHS	(S) Description
uuhost!\$*,	\$1@foo.com,		"Changing Fred's address"

The operators used in [Figure 3-7](#) are as follows:

\$* token in Pattern (LHS) resolves as anything after the exclamation point (!).

\$* = fred.jones.

The comma (,) is the separator between the LHS, RHS, and Description.

The \$1 or first token in LHS string \$* moves to the RHS as is.

Message headers are not rewritten during SMTP address name resolution. The address is parsed and rewritten by the delivery daemon rewriting rules, and passed as a logical address to the receiving daemon, which then parses and resolves it.

SMTP Server Parameters

See Also:

- ["Oracle Mail SMTP Inbound Server"](#) on page F-34 for detailed information about SMTP inbound server parameters
- ["Oracle Mail SMTP Outbound Server"](#) on page F-44 for detailed information about SMTP outbound parameters

Managing SMTP Servers

If message archiving is enabled on your Oracle Mail installation, create a server instance of the SMTP outbound server that is dedicated to the handling of archived messages.

See Also:

- ["Managing Oracle Mail Servers and Instances"](#) on page 3-2 for instructions on creating, deleting, and modifying SMTP server instances
- ["Enabling Message Archiving"](#) on page 8-29 for information about creating and configuring an SMTP outbound server instance for message archiving

SMTP Routing Control

See Also: ["Antispam Methods"](#) on page 6-5 for more information about routing control

Housekeeper Server

The **Housekeeper** server is a background process that works inside the Oracle Database and performs periodic tasks, such as garbage collection, which cleans up deleted message bodies. Additional tasks include performing Oracle Text index synchronization and optimization for enabling message body search, and moving message bodies to tertiary storage tablespaces.

During Oracle Mail installation, a Housekeeper instance is created by default to perform basic garbage collection. Once the server process finishes its assigned task, it sleeps for a configured amount of time before waking up to restart the tasks.

The Housekeeper server can also be configured to start and stop running certain tasks dynamically from the command line, enabling the administrator to control the exact time when certain tasks are started.

See Also: ["Dynamically Starting and Stopping Housekeeper Tasks"](#) on page D-22 for instructions on dynamically starting and stopping Housekeeper tasks from the command line

The Housekeeper server consumes data produced by other Oracle Mail servers. For example, the pruning garbage collection task processes all the messages produced by the IMAP `expunge` command and filters out those messages that are still referenced by users. Additionally, the `collection` task processes the rest of the pruned result and removes all message bodies. Housekeeper metric information enables administrators to track the progress of its tasks.

The Housekeeper process can also be configured to enable LogMiner-based mail recovery. Once this feature is enabled, vital information about all deleted messages is kept in the database, allowing these messages to be recovered at a later time.

Note: Enabling LogMiner recovery may lead to Housekeeper process performance degradation.

The Housekeeper log files are located in two places: the Oracle Database and the Applications tier. The log file on the Oracle Database contains information about the progress of housekeeping tasks; the log file on the Applications tier contains information about the status of the process.

The Housekeeper server is designed to perform eight distinct and separate tasks:

- Three operational tasks:
 - Statistics Cleanup
 - Process Control Message Cleanup
 - Tertiary Storage
- Five additional tasks:
 - Expiration
 - Pruning

- Collection
- Index Synchronization
- Text Optimization

This section includes the following topics:

- [Configuring Housekeeper Tasks](#)
- [Housekeeper Server Parameters](#)
- [Managing Housekeeper](#)

Configuring Housekeeper Tasks

For the Housekeeper server to perform all eight of the aforementioned tasks, either eight separate instances of the Housekeeper server can be configured, or tasks can be combined. At a minimum, three separate instances must be configured in order to perform all eight tasks.

Regardless, Oracle recommends that for any size installation, three separate instances of the Housekeeper server be configured to perform the following three tasks:

- **Collection:** Deleting an instance of a message or the message itself is an expensive operation for the database in terms of resource consumption. Therefore, for performance reasons, when you delete a message you actually tag the message to be permanently deleted later. The Housekeeper server, in the background, performs the collection and cleanup of deleted messages from the Oracle Database.

See Also: ["Process Control Message Cleanup"](#) on page 3-25 for more information about additional Housekeeper tasks that can be combined with the garbage collection instance

- **Index Synchronization:** Index synchronization is the process of looking at each message in detail (message bodies, headers, and attachments) to facilitate searching. It is another expensive operation for the database in terms of resource consumption. Therefore, for performance reasons, messages stored in the Oracle Database that must be indexed are done so by the Housekeeper server in the background.

See Also: ["Creating a Housekeeper Server Instance to Index Text"](#) on page 3-29 for more information

- **Tertiary Storage:** Users often keep messages in their production Oracle Database for long periods of time. The Oracle Database has tertiary storage, which is a separate set of tablespaces designed to store these older messages. An administrator can configure these tablespaces on less expensive or slower disks, because they will be accessed less often. The Housekeeper server, in the background, moves messages of a specified age to tablespaces in tertiary storage.

See Also: ["Tertiary Storage"](#) on page 3-30 for information about configuring a Housekeeper server instance for tertiary storage

This section includes the following topics:

- [Process Control Message Cleanup](#)
- [Oracle Text](#)

■ Tertiary Storage

Process Control Message Cleanup

As mentioned previously, a Housekeeper server instance configured for garbage collection is created by default upon installation of Oracle Mail. In addition to garbage collection, configure this instance to perform process control message cleanup.

Different Oracle Mail servers sometimes use advanced queuing to facilitate interprocess communication. The Housekeeper server can be configured to clean these messages, improving Oracle Mail performance.

Note: This procedure combines the Process Control Message Cleanup operation mode with the Pruning and Collection tasks.

To enable process control message cleanup:

1. Open the Application Server Control Console.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select the Housekeeper server and click **Stop** to bring down the server.
5. Click **Housekeeper** to display the process page.
6. Click the default Housekeeper server instance configured for garbage collection to display the parameters for that particular instance.
7. Select **Process Control Message Cleanup** from the **Operation Mode** list.
8. Select **Enabled** from the **Pruning** and **Collection** lists in the **Housekeeping Operations** section.
9. Ensure that all other Housekeeping Operations are disabled.
10. Enter 60 in the **Frequency of Execution of Housekeeper Process** field.
11. The **Age Threshold** parameter units are measured in minutes when process control message cleanup is configured. Because few of these messages are generated on a daily basis, 30 is an acceptable value for this parameter.
12. Select **On Startup** from the **Run Task** list.
13. Optionally, enable the **Support Log Miner Recovery** parameter to recover messages with LogMiner.

See Also: ["Setting Up LogMiner to Recover Mail"](#) on page 3-25

14. Click **Apply**.
15. Return to the Mail Application page.
16. Select the Housekeeper server and click **Start** to start the server.

Setting Up LogMiner to Recover Mail To set up mail recovery, you must enable supplemental logging for the Oracle Database and configure the Housekeeper server to record the messages being deleted into the redo logs.

Setting up LogMiner involves the following tasks:

- [Oracle Database Tasks](#)
- [Enabling Supplemental Logging](#)
- [Configuring the Housekeeper Server to Record Deleted Messages](#)

Oracle Database Tasks

Perform the following tasks on the Oracle Database to set up LogMiner:

1. Ensure that the database is in `archive log` mode.
2. Specify the archive log destination in the `init.ora` file, as follows:

```
log_archive_dest_1='location=full_path'
```

3. Specify the archive log format in the `init.ora` file, as follows:

```
log_archive_format=arch_%t_%s_%r.arc
```

Enabling Supplemental Logging

To enable supplemental logging for the Oracle Database:

1. Start SQL*Plus and log in as `sys`, as follows:

```
$ sqlplus sys/sys_password
```

2. Enter the following SQL command:

```
SQL > alter database add supplemental log data (primary key,unique index)
columns;
```

Configuring the Housekeeper Server to Record Deleted Messages

The following procedure is an abbreviated form of that described in "[Process Control Message Cleanup](#)" on page 3-25. The following procedure only configures the Housekeeper to record deleted messages.

To configure the Housekeeper process to record the messages being deleted into the redo logs:

1. Open the Application Server Control Console.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Click **Housekeeper**.
5. Click the Housekeeper instance where the `collection` parameter is configured.
6. Enable the **Support Log Miner Recover** parameter in the **General Parameters** section.
7. Click **Apply**.
8. Return to the Mail Application page.
9. Click **Start**.

Oracle Text

Integrating [Oracle Text](#) and Oracle Mail extends the e-mail server functionality, enabling text search in e-mails, e-mail theme generation, and e-mail formatting functions such as highlight and markup.

Oracle Text integration support is installed by default when Oracle Mail is installed. However, if the database user `ctxsys` is not present at the time of installation, the Oracle Text installation will fail.

Note: This feature is only available when the Oracle Database is Oracle Database 10g Enterprise Edition.

The user attribute **Text Indexing** enables users to perform server-side search on mail message bodies.

See Also: ["Managing Oracle Mail Users"](#) on page 2-3 to enable a user for message body search or change this attribute value for all new users created by default

Oracle Mail support for Oracle Text provides both a Java SDK and a PL/SQL SDK for application integration. Applications can interface with the SDKs to use or extend Oracle Text functionalities.

Except for zipped attachments, Oracle Mail message bodies and attachments can be indexed and later searched for text strings, themes, gists, or formatting, such as highlight and markup. To be searchable, the contents of a mail message body must be indexed by the Oracle Text server. If indexing is enabled, Oracle Mail puts candidate messages into a queue for Oracle Text to index. The created index is later usable for performing a message body search.

Text indexing enables searching message bodies for content, using IMAP clients that support message body searching or using the Oracle Ultra Search component. This feature is available only to users whose accounts are text-enabled.

Applications that integrate with Oracle Mail can use Oracle Text indexing through the PL/SQL and Java APIs.

This section includes the following topics:

- [Enabling E-mail Body Searching for Various Languages](#)
- [Verifying Oracle Text Installation](#)
- [Creating a Housekeeper Server Instance to Index Text](#)
- [Enabling Text Indexing for a User](#)

Enabling E-mail Body Searching for Various Languages The text index for e-mail body searching is created with the Oracle Text `BASIC_LEXER`, which supports English and most western European languages that use white space delimited words. For other languages that are not supported by Oracle Text `BASIC_LEXER`, such as Chinese, Japanese, and Korean, e-mail body search does not function.

To enable message search for Chinese, Japanese, or Korean:

1. Stop the Housekeeper process using the Application Server Control Console.

See Also: ["Starting, Stopping, Restarting, or Refreshing All Server Instances"](#) on page 3-5 for information about stopping a process

2. Run the following SQL commands as database user `es_mail`:

```
SQL> @recreate_text_index.sql
```

3. Select and enter the text lexer for indexing messages from the list, as shown in the following example. The script will rebuild the text index using the selected lexer.

```
SQL> @recreate_text_index
Use this script to customize or update the index used for text
searching. Choose a lexer and a default character set for mail
messages.

LEXER: a lexer is a program component responsible for recognizing
words in a document for a specific language. Choose one of the
following representing the primary language used in the system.

1 - chinese_lexer   for Chinese
2 - japanese_lexer for Japanese
3 - korean_lexer   for Korean
4 - default_lexer   for other languages
(this is the default if no input is received)

Enter the desired lexer number[4]: 4

DEFAULT CHARACTER SET: when a mail message body or a text attachment
of a mail message does not contain a character set tag in the
header, the system uses this default character set to recognize the
mail message for indexing. Choose one of the following
representing the primary character set that client mail programs
use in this environment. Note that this is NOT necessarily
the character set used in the mail database. For a list of
recommended default character sets for popular languages,
refer to the text file default_charset.txt present in the
same directory.

1 - US-ASCII (US7ASCII)           2 - UTF-8 (AL32UTF8, default)
3 - SHIFT_JIS (JA16SJIS)         4 - EUC-JP (JA16EUC)
5 - ISO-2022-JP (ISO2022-JP)
6 - EUC-KR (KO16MSWIN949)         7 - ISO-2022-KR (ISO2022-KR)
8 - GBK (ZHS16GBK)               9 - GB18030 (ZHS32GB18030)
10 - BIG5 (ZHT16MSWIN950)        11 - BIG5-HKSCS (ZHT16HKSCS)
12 - WINDOWS-1250 (EE8MSWIN1250) 13 - WINDOWS-1251 (CL8MSWIN12510)
14 - WINDOWS-1252 (WE8MSWIN1252) 15 - WINDOWS-1253 (EL8MSWIN1253)
16 - WINDOWS-1254 (TR8MSWIN1254) 17 - WINDOWS-1255 (IW8MSWIN1255)
18 - WINDOWS-1256 (AR8MSWIN1256) 19 - WINDOWS-1257 (BLT8MSWIN1257)
20 - WINDOWS-1258 (VN8MSWIN1258) 21 - TIS-620 (TH8TISASCII)

Enter the default character set number[2]: 2
Setting default character set and re-creating the text index...
(this may take a while, please wait...)
Setting default character set...
Creating text index...

PL/SQL procedure successfully completed.

Done.
SQL> exit
```

4. Ensure that a Housekeeper server instance is configured for Oracle Text indexing using the Application Server Control Console.
5. Start the Housekeeper process using the Application Server Control Console.

See Also: ["Starting, Stopping, Restarting, or Refreshing All Server Instances"](#) on page 3-5 for information about starting a process

Note: Indexing is limited to only one lexer.

Verifying Oracle Text Installation Before text indexing can be used, Oracle Text must be installed and configured. Oracle Text is installed by default when Oracle Database 10g Enterprise Edition is installed. The Oracle Mail support for Oracle Text installation fails if the database user `ctxsys` is not present at the time of installation.

To verify that the Oracle Text option was installed and configured on the Oracle Database, run the following SQL query as `sysdba`:

```
SQL> select comp_id, version, status from dba_registry;
```

If Oracle Text was installed correctly, an output similar to the following displays:

COMP_ID	VERSION	STATUS
...		
CONTEXT	10.1.0.4.0	VALID

If Oracle Text was not installed and configured on the Oracle Database, it must be configured manually.

See Also: *Oracle Collaboration Suite Installation Guide for Solaris Operating System* for further instructions on installing and configuring Oracle Text

Creating a Housekeeper Server Instance to Index Text Oracle Text periodically processes a message queue filled by a Housekeeper server instance.

Text search performance can be improved by periodically optimizing the existing Oracle Text index. Because many indexed messages are deleted or moved, the Oracle Text index bits are no longer consecutive, slowing down searching. Search time can be reduced by periodic cleanup of the Oracle Text index, removing entries that refer to deleted or moved messages.

Optimization can be done by the Housekeeper server. Assign a new Housekeeper instance to do this unless performance requires optimization to be done at the same frequency as indexing.

To create a Housekeeper server instance to queue messages for text indexing:

Note: This procedure combines the Statistics Cleanup operation mode with the Text Synchronization and Text Optimization tasks.

1. Open the Application Server Control Console.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select the Housekeeper server and click **Stop** to bring down the server.
5. Click **Housekeeper** to display the Housekeeper page.

6. Click **Create** to create a new Housekeeper server instance.
7. Click the new Housekeeper server instance to display its parameter page.
8. Select **Enabled** from the **Text Synchronization** and **Text Optimization** lists in the **Housekeeping Operations** section.
9. Ensure that all other Housekeeping Operations are disabled.
10. In the **Frequency of Execution of Housekeeper Process** field in the **General Parameters** section, enter how often the Housekeeper should queue messages for indexing, in minutes.

For example, if the Housekeeper should queue messages for indexing every 2 hours, enter 120 in the field.

Note: Indexing too frequently will result in increased database server load and degraded search performance, but indexing too infrequently will cause more new messages to be unavailable for search. The recommended frequency is 1 to 2 hours.

11. Click **Apply**.
12. Return to the Mail Application page.
13. Select the Housekeeper server and click **Start** to bring up the server.

Enabling Text Indexing for a User

See Also: ["Modifying E-mail User Attributes"](#) on page 2-6 to enable text indexing for users

Tertiary Storage

Administrators can configure Oracle Mail to move messages to **tertiary storage** based on the age of the message. This process frees up valuable space on the primary disk for newer, more frequently accessed messages, and enables users to still access messages in tertiary storage, as before.

Message stores tend to grow constantly. Mail continually enters the store, and while many messages are deleted, more are saved. Generally, older messages are accessed less, so storing them on less expensive, slower disks while keeping them accessible to users may be an acceptable way to reduce costs. Depending on the storage mechanisms used for tertiary storage, users should not be aware that their older messages have been moved to a different physical disk.

Tertiary storage in Oracle Mail is enabled through the Housekeeper. The Housekeeper can be set to move older messages to a tablespace named `ESTERSTORE`, which is reserved for tertiary storage of old messages. The age of messages to be moved to tertiary storage is set through the **Age Threshold** Housekeeper general parameter.

Note: For the name of mail store tablespaces and their default storage parameters, refer to the `$ORACLE_HOME/oes/install/sql/tblspc.sql` script.

Tertiary storage can be initially planned as part of an Oracle Mail system, or it can be implemented later. By default, the `ESTERSTORE` tablespace is created on the same disk as all other tablespaces during the initial Oracle Mail installation.

[Table 3–4](#) gives the four considerations in determining how tertiary storage tablespace is handled.

Table 3–4 Tertiary Storage Usage and the ESTERSTORE Tablespace

If Tertiary Storage Is:	Then ESTERSTORE Tablespace:
Never enabled for the system	Remains empty.
Enabled for Oracle Mail	Is set up on a disk different from the primary mail store, either before or after Oracle Mail installation.
To be implemented for a new Oracle Mail system	Is created on a disk different from the primary mail store, before installation of Oracle Mail.
To be implemented for an existing Oracle Mail system	Is moved, with the <code>es_tbody</code> table, from its default location on the same disk as the primary mail store onto a separate disk.

See Also: *Oracle Database Administrator's Guide* for more information about creating and moving tablespaces

This section includes the following topics:

- [Moving the ESTERSTORE Tablespace](#)
- [Enabling Tertiary Storage](#)

Moving the ESTERSTORE Tablespace To move the ESTERSTORE tablespace after Oracle Mail has already been installed:

1. Back up the database.
2. Identify the data files for ESTERSTORE tablespace.

For example, the following query of the data dictionary view `DBA_DATA_FILES` lists the data file names of the ESTERSTORE tablespace:

```
select file_name from dba_data_files
where tablespace_name='ESTERSTORE';
FILE_NAME
-----
/usr/app/oracle/product/mailstore/dbf/erstore.dbf
```

3. Take the ESTERSTORE tablespace offline.
4. Copy the data files for the ESTERSTORE tablespace using the operating system, to a different disk.
5. Use the `ALTER TABLESPACE` statement with the `RENAME DATAFILE` clause to change the file names for the ESTERSTORE tablespace to a new location.

```
alter tablespace esterstore rename datafile
'/usr/app/oracle/product/mailstore/dbf/erstore.dbf' to
'file_name_in_new_location';
```

6. Bring the ESTERSTORE tablespace online.

```
alter tablespace esterstore online;
```

See Also: *Oracle Database Administrator's Guide*

Enabling Tertiary Storage After the `ESTERSTORE` tablespace has been created, create and configure an instance of the Housekeeper server to enable tertiary storage:

Note: This procedure combines the Tertiary Storage operation mode with the Expiration task.

1. Open the Application Server Control Console.

See Also: ["Oracle Enterprise Manager Application Server Control Console"](#) on page 1-3 for information about accessing the Application Server Control Console

2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Select the Housekeeper server and click **Stop** to bring down the server.
5. Click **Housekeeper** to display the Housekeeper page.
6. Click **Create** to create a new Oracle Mail Housekeeper instance.
7. Click the new Housekeeper instance to display its parameter page.
8. Select `Tertiary Storage` from the **Operation Mode** list in the **Housekeeping Operations** section.
9. Select `Enabled` from the **Expiration** list in the **Housekeeping Operations** section.
When **Expiration** is enabled, each folder deletes messages older than the number of days specified in the **Age Threshold** parameter.
10. Ensure all other Housekeeping Operations are disabled.
11. In the **Frequency of Execution of Housekeeper Process** field, enter how often the Housekeeper should perform tertiary storage, in minutes.
For example, if the Housekeeper process performs tertiary storage once a day, enter 1440 (24*60).
12. In the **Age Threshold** field, enter the age, in days, of messages you want to move to tertiary storage. The default is 30 days.
For example, if you enter 60 in this field, messages that are 60 days old are moved to tertiary storage.
13. Select `On Startup` from the **Run Task** list.
14. Click **Apply**.
15. Return to the Mail Application page.
16. Select the Housekeeper server and click **Start** to bring up the server.
The Housekeeper process periodically moves messages of the appropriate age into tertiary storage.

Housekeeper Server Parameters

See Also: ["Oracle Mail Housekeeper"](#) on page F-2 for detailed information about Housekeeper parameters

Managing Housekeeper

See Also: ["Managing Oracle Mail Servers and Instances"](#) on page 3-2 for instructions on creating, deleting, and modifying Housekeeper instances

List Server

List servers enable public distribution list management as well as integration with other messaging services or applications.

Users can own and administer public distribution lists as a way to distribute information to groups of people or as a discussion forum. If desired, restrictions can be placed on membership, requiring prior approval, and on outgoing messages, requiring screening by one or more moderators who control what messages are sent out. For example, a distribution list administrator may screen out advertisements.

The **List Server** is installed with Oracle Mail, with default values set for all List Server parameters. Administrators can modify these values to meet performance or feature requirements. For example, a distribution list with a large number of members requires changing the Oracle Internet Directory Query Entry Return Limit parameter. It must be configured to return a large number of entries to enable the list resolution API to return all the members. This parameter can be configured through the Oracle Internet Directory administration tool **oidadmin**.

See Also: Chapter 4, "Directory Administration Tools" in *Oracle Internet Directory Administrator's Guide* for information about how to access **oidadmin**

APIs provided with the List Server enable users to customize distribution lists and messages sent out to a distribution list. For example, marketing campaigns can send special nontransferable offers readable only by the intended recipients. As another example, a user can query a sales information database to create a distribution list of all customers who have made purchases in the past three months. Customers on that distribution list can then receive e-mail coupons with discounts based on the amount of their purchases.

This section includes the following topics:

- [List Server Mail Interface](#)
- [Archiving Distribution List Posts](#)
- [External Distribution Lists](#)
- [Distribution List Digests](#)
- [Distribution List Bounce Processing](#)
- [Multiple Language Responses](#)
- [Mail Merge](#)
- [List Server Parameters](#)
- [Managing List Servers](#)

List Server Mail Interface

The List Server features a mail interface that enables distribution list members and owners to perform certain tasks. Depending on the distribution list type and

parameters, members and owners can subscribe, unsubscribe, suspend, resume, or invite members to a distribution list. The command messages are sent to a particular e-mail address of the form `list_name-admin@domain`. For example, if a distribution list name is `list@foo.com`, commands are sent to `list-admin@foo.com`.

See Also: Oracle WebMail online Help for a complete list of commands with syntax, description, and examples

Archiving Distribution List Posts

A distribution list owner can have all posts sent to a distribution list stored as messages in an archive in the NNTP server archives. Such an archive operates as a newsgroup and can be browsed using a standard news client.

An administrator must specify archiving as a property of the distribution list in order to archive messages, as described in [Table 2-4, "Distribution List Parameters"](#) on page 2-16.

When a distribution list is archived, a newsgroup is created with a name reflecting the original distribution list. For example, the name of the NNTP archive newsgroup for the distribution list `abc@foo.com` becomes the following:

```
ListArchive.abc
```

Once a distribution list is created, the domain administrator can begin archiving, which affects only e-mails sent after the archive property is set for a distribution list. No messages prior to that time are archived.

Distribution list archives must have the post parameter disabled. An e-mail is added to the archive only when an e-mail is delivered to the distribution list. E-mails cannot be added to an archive by any other mechanism.

Distribution list archives must be local to the domain of the distribution list. Global newsgroups cannot be associated with a distribution list as an archive.

Administrators can set expiration periods for distribution list archives, such as one month, meaning that messages are only stored in the archive for one month before being deleted. The expiration policy for a distribution list's archive is the corresponding newsgroup's expiry attributes.

External Distribution Lists

External distribution lists provide a way for the membership of a distribution list to be stored outside of Oracle Mail, while using the List Server to deliver e-mails to such a distribution list. A distribution list owner or domain administrator can configure a distribution list to be external by enabling the external list option in the distribution list properties page, as described in [Table 2-4, "Distribution List Parameters"](#) on page 2-16. A PL/SQL procedure for resolving the addresses of the distribution list members must be created on the Oracle Database to which the List Server is connected.

The PL/SQL procedure must have the following syntax:

Note: Database links are not supported for any of the List Server PL/SQL callouts (either for PL/SQL mail-merge or for external distribution lists). The procedures for both of these features must reside on the queue database (the database to which the List Server is connected). These procedures can then, in turn, refer to procedures on other databases using database links. The server itself does not support database links.

```
procname(session_ID IN NUMBER,
  msg_obj IN MAIL_MESSAGE_OBJ,
  list_ID IN VARCHAR2,
  return_count IN NUMBER,
  count OUT NUMBER,
  recipients OUT RECIPIENTS_TABLE)
```

- The `session_ID` and `msg_obj` parameters can be used by the PL/SQL procedure developer to call the `MAIL_MESSAGE` PL/SQL API and fetch any relevant information about the message being processed.

See Also: "MAIL_MESSAGE Package" in Chapter 1 of *Oracle Mail Application Developer's Guide* for more information about the `MAIL_MESSAGE` package

- `list_ID` is the e-mail address of the list being resolved.
- `return_count` indicates whether the procedure should return the number of recipients in the list or the recipients itself. A value of 1 indicates the count is to be returned and 0 indicates that the recipients are to be returned.
- The count of the recipients are returned in the `count` variable if the value of `return_count` is 1.
- The recipients are returned in the `recipients` table with one row for each recipient if the value of `return_count` is 0.

The List Server calls the procedure twice while resolving an external distribution list. Initially, the value passed for the `return_count` parameter is 1. It receives, in return, the `count OUT NUMBER` parameter, containing just the number of recipients in the list, and not the list of recipient addresses.

Subsequently, the value passed for the `return_count` parameter is 0, which causes the procedure to return a table of recipients in the `recipients OUT` parameter. Each row of that table contains the full e-mail address of a recipient.

Example

The following is an example of how to create the `get_cust_list` PL/SQL procedure.

1. Open the Application Server Control Console.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Click **List Server** to display the List Server page.
5. Navigate to the list `cust_list@acme.com`.

6. Edit the properties of the list.
7. Select **True** in the list box for the **Group Is External** field.
8. Set the **External Procedure** parameter to `get_cust_list`.
9. Connect as `es_mail` to the Oracle Database to which the List Server is connected and create the following PL/SQL procedure:

```
procedure get_cust_list(session_id IN NUMBER,
    msg_obj IN MAIL_MESSAGE_OBJ,
    listid IN VARCHAR2,
    return_count IN NUMBER,
    count OUT NUMBER,
    recipients OUT RECIPIENTS_TABLE)
l_hdr varchar2(2000);
l_cnt number;
begin
    -- First get the sender of the message being processed.

    mail_message.get_header(session_id, msg_obj, 'From', l_hdr);

    -- Query the table to get the customer count based on the sender's region.

    if (instr(lower(l_hdr), 'sales_us@foo.com', 1, 1) > 0)
        then select count(*) into l_cnt from customer_list where region='US';

        else
            select count(*) into l_cnt from customer_list where region!='US';
        end if;

    -- If the count is requested, then return the count.

    if (return_count = 1)
        then count := l_cnt;

        else

    -- Create a recipients_table object.

        recipients := recipients_table();
        recipients.extend(l_cnt);

    -- Query the table to get the customer list based on the sender's region.

        if (instr(lower(l_hdr), 'sales_us@foo.com', 1, 1) > 0)
            then select mailid from customer_list where region='US'
                bulk collect into recipients;

            else
                select mailid from customer_list where region!='US'
                bulk collect into recipients;

            end if;
        end if;
    end;
```

The previous example assumes that `cust_list@acme.com` is a list of customers maintained in a database table by a different application. This procedure uses a table called `customer_list`, described in [Table 3-5](#).

Assuming that the `customer_list` table is populated with the customer e-mail IDs and their regions, if the user `sales_us@foo.com` sends a message to the list, the message is delivered to all the customers in the `us` region. Mail from any other e-mail address to this list is delivered to all customers on the list, regardless of region.

Table 3–5 *customer_list Table*

Column Name	Data Type	Explanation
<code>mailid</code>	<code>varchar2 (256)</code>	E-mail ID of a customer
<code>region</code>	<code>varchar2 (256)</code>	Geographical region of the customer

Distribution List Digests

The List Server includes a mail digest feature for distribution lists. This feature enables a member of a distribution list to receive a single message containing all posts in a day or a week to a specific distribution list, rather than receive the individual messages as they are sent.

Distribution list owners control whether digests are enabled on the lists they own or not. If digests are enabled for a distribution list, a member can choose to receive a daily or a weekly digest, by changing the setting through the Oracle WebMail client or by setting the `frequency` option in the `subscribe` command (described in "[List Server Mail Interface](#)" on page 3-33).

All mail digests are sent in HTML format with a table of contents listing all the postings in that digest. Users can navigate to the specific post by clicking the subject of the post in the table of contents.

If the distribution list is also configured for archiving, a member can choose not to receive any messages. The user will still be able to browse through the archives of the distribution list and not receive any of the messages sent to the list. By default, a user will receive all messages sent to the distribution list as they are delivered.

Distribution List Bounce Processing

It is possible that certain members of a distribution list (especially those in different domains) are invalid or do not exist. In such cases, a DSN is generated when a message is sent to one of these users and this DSN is sent to the sender of the message.

The List Server can be configured to automatically handle these DSNs. If the distribution list owner enables bounce processing for a list (using the Oracle WebMail client), the List Server receives all DSNs and a record is maintained.

Once the number of bounces for a particular e-mail address reaches a preconfigured threshold, the owner of the distribution list is notified of this, along with the action the owner can take to remove the invalid recipient mail ID from the list. The distribution list owner can then unsubscribe the user from the list.

Multiple Language Responses

With Oracle Mail, all responses sent out by the List Server to users (in response to commands or requests for message moderation, for example), are delivered in the preferred language of the recipient.

Mail Merge

Mail merge enables customized mail to be delivered to every member of a distribution list. Distribution list owners or domain administrators must decide on a mail merge

tag for a list and set it in the distribution list properties page. The mail merge tag can be a single word or a group of words. This feature can be enabled for a distribution list by providing a value for the merge tag property of the list, as described in [Table 2-4, "Distribution List Parameters"](#) on page 2-16. [Table 3-6](#) lists the two types of mail merge that the List Server supports.

Table 3-6 Types of Mail Merge and Customizable Features

Type of Mail Merge	Description	Customizable Features
Standard mail merge	Message contents can be customized for each recipient with the values in the Customizable Features column.	Recipient's mail address (recipient_mail_address) Recipient's first name (recipient_first_name) Recipient's last name (recipient_last_name) Recipient's full name (recipient_full_name) Current date (current_date) Current time (current_time)
PL/SQL mail merge	Similarly customizable, but also enables embedding of PL/SQL in messages. (The PL/SQL function must return a varchar2 string.)	For each recipient, the PL/SQL function is executed and the output is embedded in the mail before delivery. Any parameter defined for standard mail merge can be included as a parameter to the PL/SQL function.

For standard mail merge, use the mail merge tag appropriate to a corresponding section of the mail. For example, if the distribution list's mail merge property is `orcl`, and the mail is addressed with the recipient's full name, the mail looks like the following:

```
Dear <orcl>recipient_full_name</orcl>,
...
...
```

For PL/SQL mail merge, if you have a PL/SQL `getSalary` function that returns an individual's salary, given his mail address, you can use it in the mail. For example, you can embed the function call in the mail you send to a list of employees, letting them know their salaries, as follows:

```
Dear <orcl>recipient_full_name</orcl>,
Your salary is <orcl>getSalary(recipient_mail_address)</orcl>.
...
```

Note: Database links are not supported for any of the List Server PL/SQL callouts (either for PL/SQL mail-merge or for external distribution lists). The procedures for both of these features must reside on the queue database (the database to which the list server is connected). These procedures can then, in turn, refer to procedures on other databases using database links. The server itself does not support database links.

```
Dear <orcl>recipient_full_name</orcl>,
Your salary is <orcl>getSalary(recipient_mail_address)@dblink</orcl>.
...
```

Example

Note: The following example assumes that distribution lists and users have been set up correctly with the List Server process configured and running.

The following example shows how to create the `get_sal` PL/SQL procedure:

1. Open the Application Server Control Console.
2. Click the application server instance where Oracle Mail is installed.
3. Click **Mail Application** in the **System Components** section to display the Mail Application page.
4. Navigate to the list `all_emp@acme.com`.
5. Edit the properties of the list and set the `Group merge tag` parameter to `mail merge`.
6. Connect as `es_mail` to the mail store supported by the List Server and create the following PL/SQL procedure:

```
CREATE OR REPLACE FUNCTION get_sal(email IN VARCHAR2) RETURN VARCHAR2
mon varchar2(10);
tmp number;
ret varchar2(4000);
begin
  -- get the month and salary value for the user
  select month, salary into mon, tmp from emp_payroll where employee=email;

  -- concatenate to form a string
  ret := mon || ' is $' || tmp;

  return ret;
end;
```

The procedure assumes that some application puts employee payroll information into a database table. [Table 3-7](#) lists the columns contained in the `emp_payroll` table in the previous example.

Table 3-7 Example emp_payroll Table

Column Name	Data Type	Explanation
employee	varchar2(1000)	Mail ID of an employee
month	varchar2(10)	Month for which the salary is stored
salary	number	Salary of the employee

Following is an example of a message with mail merge tags embedded in it. This sends a mail to each recipient in the distribution list `all_emp@acme.com` with each person's salary details.

Dear <mailmerge>recipient_full_name</mailmerge>,

Your salary for the month of <mailmerge>get_sal(recipient_mail_address)</mailmerge> been credited to your account.

Thanks

Payroll

Scheduled Mail Delivery

Scheduled mail delivery enables administrators to schedule mail delivery to occur at a particular time, such as during low traffic hours, possibly minimizing server loads during peak usage hours. Otherwise, delivery of very large messages or of mailings to distribution lists with large numbers of subscribers can degrade performance.

This feature can be enabled by providing a value for the mail merge property of the distribution list. Specify the delivery time for a message by putting the schedule mail delivery tag anywhere in the mail. The following example illustrates this, using `orcl` as the tag for the mail merge property of the distribution list:

```
<orcl>send_schedule=DD-MON-YYYY hh24:mi [+/-TZH:TZM]</orcl>
```

```
<orcl>send_schedule=23-JUN-2003 21:45 -08:00</orcl>
```

Note: +/- before TZH:TZM is required.

Table 3–8 lists parameter descriptions for the `send_schedule` delivery tag.

Table 3–8 *send_schedule Parameter Descriptions*

Parameter	Description
DD	The date
MON	The three letter abbreviation for the month
YYYY	The year
hh24	The time in a twenty-four hour period
mi	The time in minutes
TZH	The optional time zone hour offset
TZM	The optional time zone minute offset

If TZH and TZM are not specified, the List Server uses the sender's time zone to schedule delivery of the mail.

List Server Parameters

See Also: ["Oracle Mail List Server"](#) on page F-25 for detailed information about List Server parameters

Managing List Servers

See Also: ["Managing Oracle Mail Servers and Instances"](#) on page 3-2 for instructions on creating, deleting, and modifying List Server instances

NNTP Server

Network News Transport Protocol ([NNTP](#)) is used to distribute, query, post, and retrieve news articles from the Internet using a reliable stream-based mechanism. NNTP enables news-reading clients to select news articles from a central database,

enabling subscribers to retrieve only the articles they want to read. The net news model provides indexing, cross-referencing, and message expiration. For server-to-server interaction, NNTP is designed to efficiently transmit net news articles over a reliable communication channel. Receiving and sending news articles is an interactive mechanism so that articles already present are not retransmitted.

The Oracle Mail NNTP server can be used immediately upon installation and configuration. In addition, the NNTP server uses the Oracle Database as an article repository and the Infrastructure directory service to store operational parameters. All protocol exchanges are performed over a stream-based connection.

During installation, default values are set for all Oracle Mail NNTP server parameters. Administrators can modify these values to meet performance or feature requirements of their site.

This section includes the following topics:

- [About News Servers](#)
- [Controlling News Storage](#)
- [Article Caching for Performance](#)
- [NNTP Processes](#)
- [NNTP Server Parameters](#)
- [Managing NNTP Servers](#)

About News Servers

One or more news servers used by the same community of users is called a **news site**. Such sites can exchange news articles, transmitting locally posted articles to other sites to provide (and serve) a wider audience. News servers that exchange news articles are called **peers**.

News articles collected into similar-topic groupings are called **newsgroups**, such as articles about sailing or articles about Oracle Database. A peer can be configured to download articles only for particular newsgroups.

Users of news services exchange information by posting and reading news messages. Posting means a news user composes a message in a standard newsreader and sends it to the news server for storage, after which other users can read it.

The NNTP service maintains a list of peer servers, the newsgroups each peer server is configured to receive, and a list of newsgroups that the NNTP service delivers. The administrator for each newsgroup specifies the peers to be fed articles from that newsgroup. Once peers and newsgroups are configured and the feed rules are set, the service is ready for posting, reading, and feeding news.

The NNTP inbound server receives news articles from clients and posts them to the Oracle Database.

The NNTP outbound server takes articles from the Oracle Database and exchanges those articles with various peers.

Controlling News Storage

News articles are stored in the Oracle Database. Inbound and outbound servers connected to the Oracle Database only handle newsgroups created in that database. News articles automatically expire in strict accordance with the Housekeeper process.

The NNTP service also creates a history record to track articles already received for which new entries are rejected by the server. Having a long expiry history avoids repeated entry of the same article into the mail store. History records are also subject to expiry and collection by the Housekeeper service.

The storage needed for NNTP service depends on the volume of incoming traffic and the expiry policy of the server instances, such as how long articles are retained. Each article's expiration date and history record are established when it is stored, and cannot later be changed.

Article Caching for Performance

You can configure the NNTP inbound server to cache articles in memory, improving performance for articles that are requested repeatedly, because no new Oracle Database access is needed. Caching can only be done for articles less than 4 KB in size. You can adjust the total cache size to accommodate the number of articles you want to cache, to provide newsreaders with quicker response times for popular articles.

NNTP Processes

There are two types of newsgroup exchanges, which are known as feeds:

- **Inbound:** The NNTP server receives the feed
- **Outbound:** The NNTP server transmits the feed

The processes required for NNTP service are shared between the inbound and outbound servers.

This section includes the following topics:

- [NNTP Inbound Server](#)
- [NNTP Outbound Server](#)

NNTP Inbound Server

The NNTP inbound server has two functions:

- **Incoming feed:** accepts news articles and waits for connections from remote hosts
- **Read or post:** accepts and waits for connections from newsreading clients and enables them to post or read news articles

The NNTP inbound server identifies the connecting host for each connection it receives. For a peer, the server prepares to receive the news feed. If the connecting host is not a peer, it is a newsreader and only has permission to read and post articles.

When a newsgroup is configured, administrators can specify the number of peers that must be sent articles for that newsgroup. Based on the number of peers specified, the NNTP inbound server creates queues of incoming messages that must be passed on to other peers.

NNTP Outbound Server

The NNTP outbound server periodically connects to peer news servers configured to receive news feeds and offers a list of new articles queued for sending. Peer server parameters determine what is offered and what is acceptable or rejected by the peer.

See Also: ["Managing Peer Servers"](#) on page 2-24 for more information

The NNTP outbound server maintains a list specifying the newsgroups for each peer server. When the outbound server contacts a peer and provides a list of new articles, the remote host's response determines which articles are sent.

NNTP Server Parameters

See Also: ["Oracle Mail NNTP Inbound Server"](#) on page F-12 and ["Oracle Mail NNTP Outbound Server"](#) on page F-15 for detailed information about NNTP server parameters

Managing NNTP Servers

See Also: ["Managing Oracle Mail Servers and Instances"](#) on page 3-2 for instructions on creating, deleting, and modifying NNTP server instances

